



KTU NOTES

The learning companion.

**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPERS**

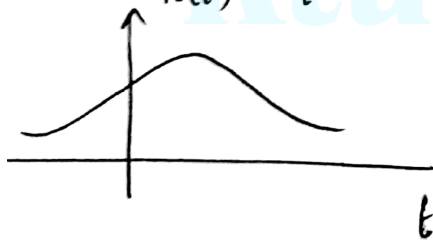
Website: www.ktunotes.in

Module - 4

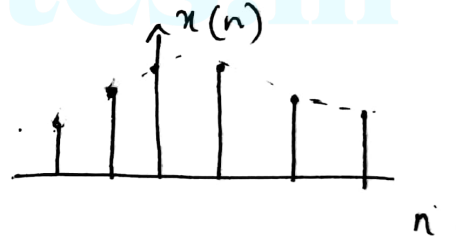
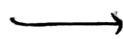
Sampled Data Systems & Z-Transform.

Sampling process

Sampling is the conversion of a continuous time signal (or analog signal) into a discrete time signal without any loss of information.



Cont. time signal



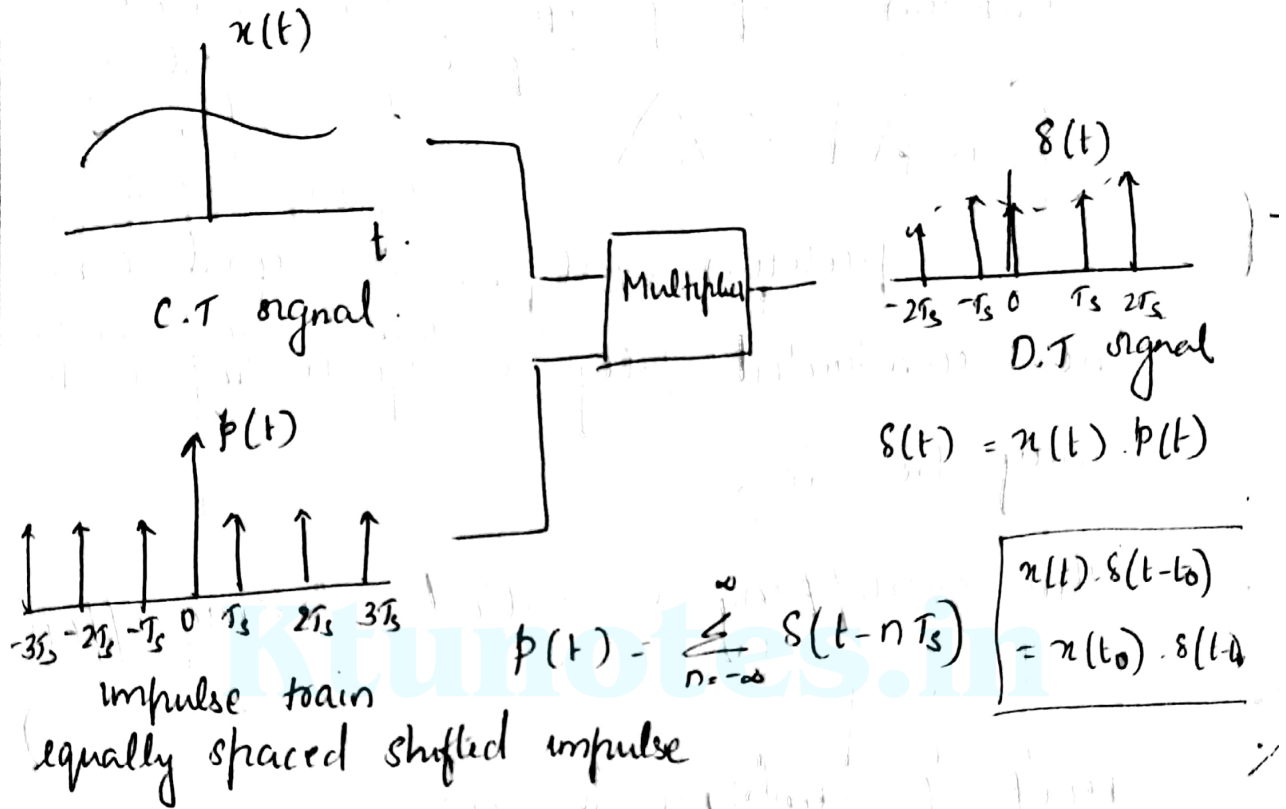
discrete time signal

Sampling theorem.

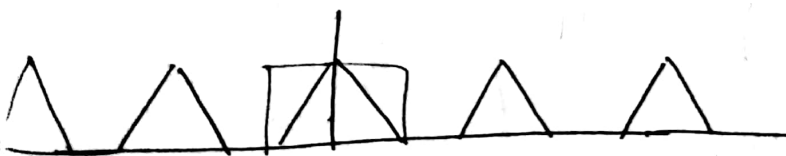
Reconstructed $\rightarrow$ DT to CT
If a signal ~~is~~ ^{can} be reconstructed only if sampling frequency is greater than or equal to twice the max. freq.

$$f_s \geq 2f_m$$

f_s - sampling freq.
 f_m - max. freq.



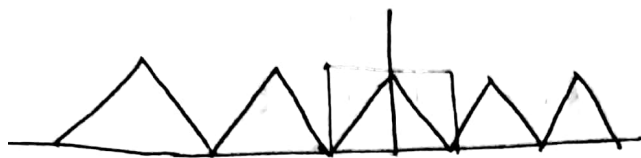
C.T signal multiplied with impulse train
 output will be D.T signal - sampling process.



Spectrum.

over sampling
 $f_s \geq 2f_m$

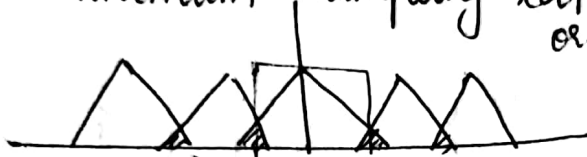
Signal reconstructed
 using low pass filter



perfect sampling
 $f_s = 2f_m$

Nyquist rate.

Minimum sampling rate reqd to recover the original signal is called nyquist rate.



$f_s \leq 2f_m$

under sampling

spectrum overlaps, so signal cannot be reconstructed fully, as original signal cannot be recovered as signals information may be loss.

This overlapping is called aliasing effect.

Proof of sampling Theorem

Impulse train sampling

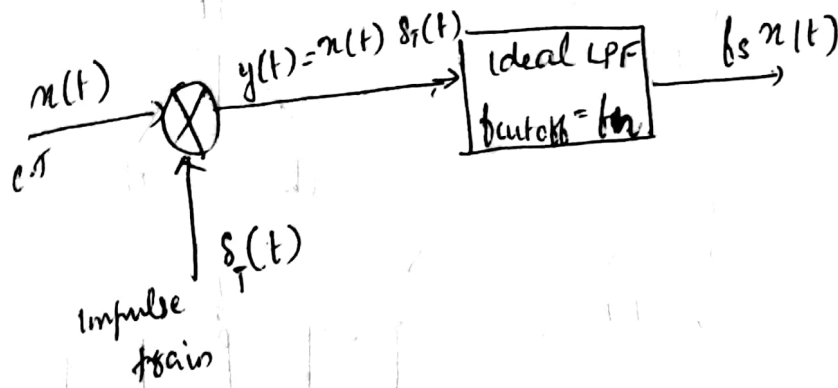
Ideal sampling

Uniform sampling

Instantaneous sampling

diff. names of
sampling process

→ Show that the spectrum of the sampled signal is the infinite sum of shifted replicas of the spectrum of original signal.



$x(t)$ is continuous time signal

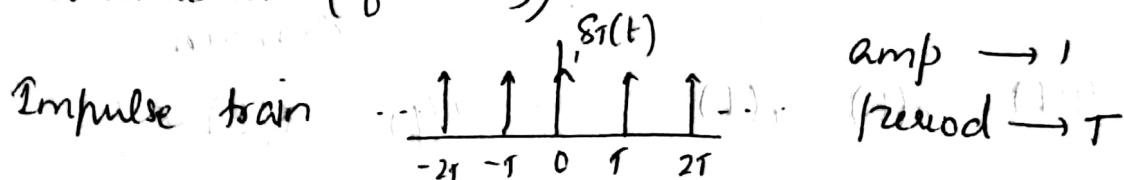
$s_T(t)$ is impulse train

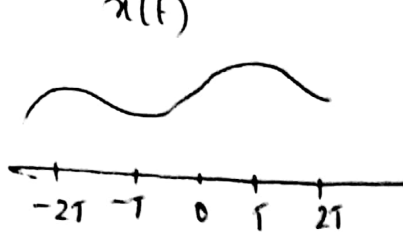
When both are multiplied $y(t) = x(t)s_T(t)$

a discrete signal is obtained.

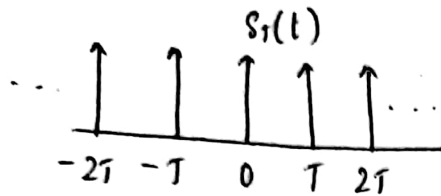
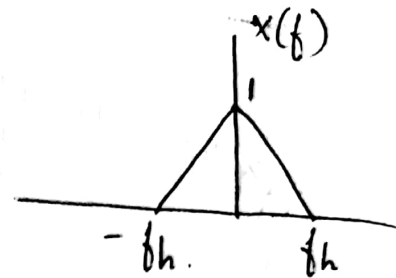
When this DT signal is passed through low pass filter (LPF) with cut off freq. = f_h

(highest freq.), the original signal can be reconstructed ($f_s \cdot x(t)$).

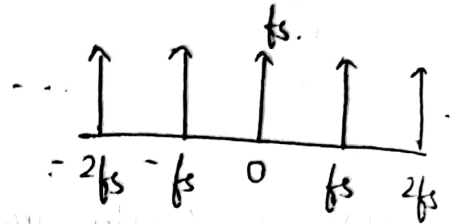




Fourier
transform



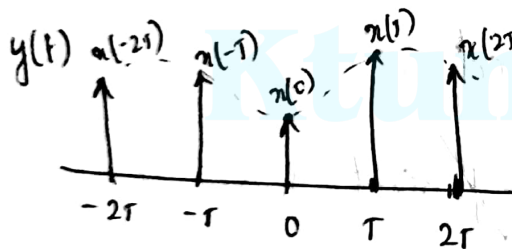
$\longleftrightarrow$ FT



$$\delta_T(t) \xleftrightarrow{FT} fs \sum_{n=-\infty}^{\infty} \delta(f - nfs)$$

multiply $x(t)$ & $\delta_T(t)$

where $fs = \frac{1}{T}$



$y(t)$ contains $x(t)$.

extract $x(t)$ using LPF

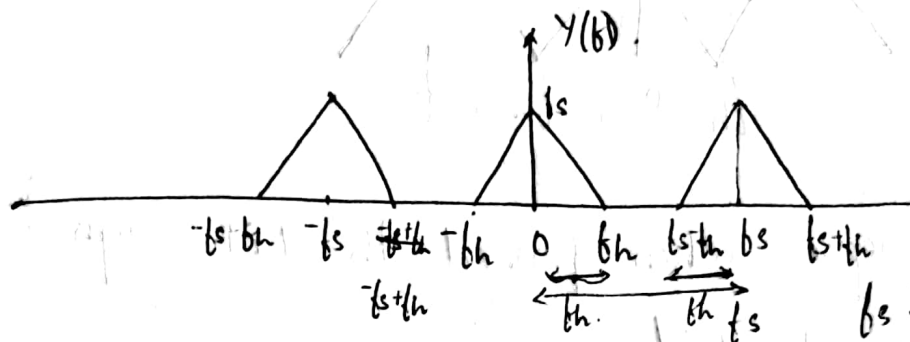
Time domain multiplication = convolution in freq. domain.

$$y(t) = x(t) \cdot \delta_T(t) \longrightarrow Y(f) = X(f) * fs \sum_{n=-\infty}^{\infty} \delta(f - nfs)$$

$$= fs \sum_{n=-\infty}^{\infty} \delta(f - nfs) * X(f)$$

$$Y(f) = fs \sum_{n=-\infty}^{\infty} X(f - nfs)$$

$$Y(f) = f_s \left[x(f) + x(f - f_s) + x(f - 2f_s) + \dots \right. \\ \left. + x(f + f_s) + x(f + 2f_s) + \dots \right]$$



$$\begin{aligned} f - f_s &= 0 \\ f &= f_s \\ f + f_s &= 0 \\ f &= -f_s \end{aligned}$$

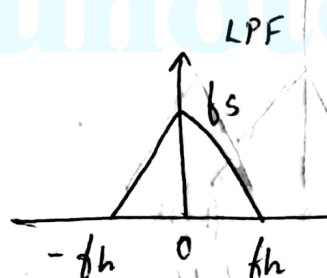
freq. spectrum

$$f_s > 2f_h$$

over sampling

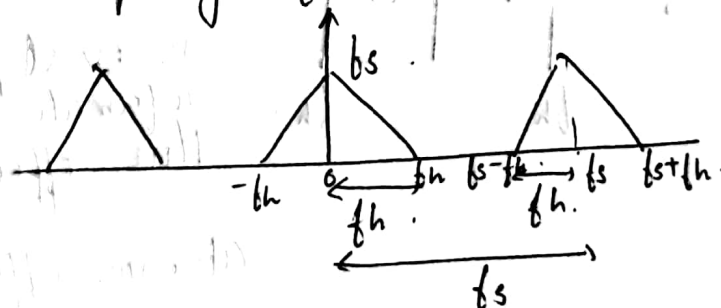
This signal is passed through LPF. with

cut-off freq. = f_h .



$f_s x(f) \rightarrow$ CT signal.

is (1) Over sampling $f_s > 2f_h$.

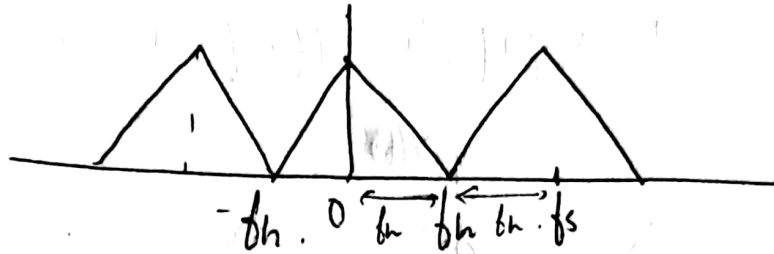


$$f_s > 2f_h$$

if passed th' LPF original signal can be recovered

Case (2)

$f_s = 2f_h$. critical sampling



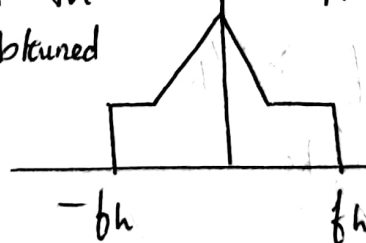
This signal when passed th' LPF signal can be reconstructed.

Case (3)

$f_s < 2f_h$. under sampling



When passed th' LPF, signal obtained is



so signal cannot be reconstructed as there is overlapping and thus information is lost. This effect is called

aliasing effect

Nyquist rate and Nyquist interval

$$\rightarrow x(t) = 2 \cos(200\pi t) + 3 \sin(600\pi t) + 8 \cos(1200\pi t)$$

Find Nyquist rate and nyquist interval.

$$a: 2\pi f = \omega$$

$$2\pi f_1 = 200\pi \Rightarrow f_1 = 100$$

$$2\pi f_2 = 600\pi \Rightarrow f_2 = 300$$

$$2\pi f_3 = 1200\pi \Rightarrow f_3 = 600$$

Take the max.
freq.

$$\text{Nyquist rate} = 2 f_{\text{max}}$$

$$= 2 \times 600 = \underline{\underline{1200 \text{ Hz}}}$$

$$\text{Nyquist interval} = \frac{1}{\text{Nyquist rate}} = \frac{1}{\underline{\underline{1200}}} \text{ s}$$

→ Determine Nyquist sampling rate & sampling interval

$$x(t) = \frac{1}{2} \sin(100\pi t) + \frac{1}{3} \sin(50\pi t)$$

$$A: 2\pi f = \omega$$

$$\left. \begin{aligned} 2\pi f_1 &= 100\pi \Rightarrow f_1 = 50 \\ 2\pi f_2 &= 50\pi \Rightarrow f_2 = 25 \end{aligned} \right\} f_m = 50$$

$$\text{Nyquist rate} = 2f_m = 2 \times 50 = \underline{\underline{100 \text{ Hz}}}$$

$$\text{Nyquist interval} = \frac{1}{\text{Nyquist rate}} = \frac{1}{100} = \underline{\underline{10 \text{ ms}}}$$

$$\rightarrow \sin^2 200\pi t$$

$$2\pi f = 2 \times 200\pi \rightarrow \text{square} \rightarrow \text{multiply by 2} \\ \text{freq. is double.}$$

$$f = 200$$

$$\text{Nyquist rate, } f_s = 2f = \underline{\underline{400 \text{ Hz}}}$$

$$\rightarrow \sin^3 100\pi t$$

$$2\pi f = 3 \times 100\pi \rightarrow f = \frac{300}{2}$$

$$\text{Nyquist rate, } f_s = 2f = 2 \times \frac{300}{2} = 300$$

$$x(t) = \sin 100\pi t \cdot \cos \frac{200\pi t}{2}$$

For multiplication $\rightarrow$ add the freq.

highest freq. is sum of individual freq.

$$f_m = 100 + 200 = \underline{\underline{300}}$$

OR

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B)$$

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$\begin{aligned} \sin 100\pi t \cdot \cos 200\pi t &= \frac{1}{2} [\sin(100\pi + 200\pi)t + \sin(100\pi - 200\pi)t] \\ &= \frac{1}{2} [\sin(300\pi t) - \sin 100\pi t] \end{aligned}$$

$$2\pi f_1 = 300\pi \quad \rightarrow \quad f_1 = \frac{300}{2} = 150$$

$$2\pi f_2 = 100\pi \quad \rightarrow \quad f_2 = \frac{100}{2} = 50$$

$$\text{Nyquist rate, } f_s = 2f_m = 2 \times 150 = \underline{\underline{300 \text{ Hz}}}$$

$$\rightarrow x(t) = \text{sinc } 100t$$

$$\text{sinc } t = \frac{\sin \pi t}{\pi t}$$

$$\text{sinc } 100t = \frac{\sin \pi t}{100\pi t}$$

$$2\pi f = 100\pi \Rightarrow f = 50$$

$$\text{N. rate, } f_s = 2 \times 50 = \underline{\underline{100 \text{ Hz}}}$$

$$\rightarrow x(t) = \text{sinc}^2 100t$$

$$\text{sinc}^2 100t = \left(\frac{\sin \pi t}{100\pi t} \right)^2$$

$$2\pi f = 200\pi \Rightarrow f = 100$$

$$\text{N.R } f_s = \underline{\underline{200 \text{ Hz}}}$$

→ determine Nyquist rate of sampling for signals

i) $x(t) = 2 \sin 2500t + 3 \cos^2 500t$

ii) $x(t) = 10 \sin 500t$

a.) $x(t) = 2 \sin 2500t + 3 \cos^2 500t$

$$2\pi f_1 = 2500 \Rightarrow f_1 = \frac{2500}{2} = 125$$

$$2\pi f_2 = 1000 \Rightarrow f_2 = \frac{1000}{2\pi} = 159.2$$

$$\text{N.R.}, f_s = 2f_m = 2 \times 159.2 = \underline{\underline{318.4 \text{ Hz}}}$$

ii) $x(t) = 10 \sin 500t$

$$= 10 \frac{\sin 500t}{500t}$$

$$2\pi f = 500 \Rightarrow f = \frac{500}{2\pi} = \frac{500}{2}$$

$$\text{N.R.}, f_s = 2 \times \frac{500}{2} = \underline{\underline{500}}$$

$$\rightarrow x(t) = 2 \cos 400\pi t + 6 \cos 600\pi t$$

$$2\pi f_1 = 400\pi$$

$$f_1 = 200$$

Take the max. freq.

$$2\pi f_2 = 600\pi$$

$$f_2 = 300$$

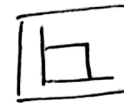
$$N.R., f_s = 2 \times 300 = \underline{\underline{600}}$$

Sample and hold

Zero order hold (ZOH)



sampled signal.



hold.



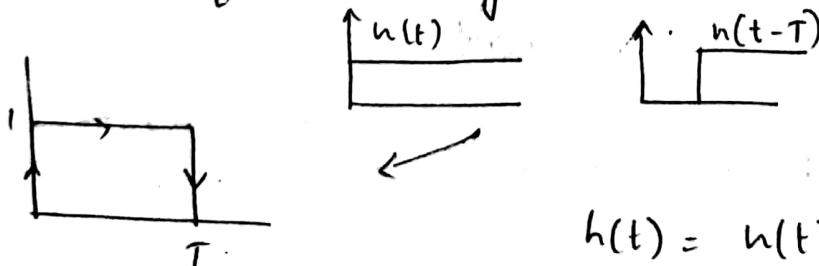
ZOH.

It is a filter or interpolator.

ZOH is a common tool to convert D.T signal back to C.T. Whenever a sample enters into ZOH system,

those samples are held for a certain

duration of time by ZOH.

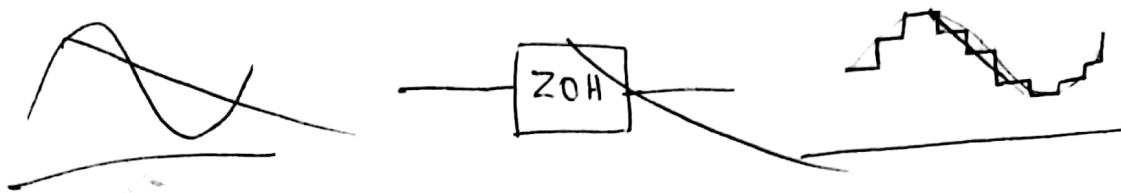


$$h(t) = u(t) - u(t-T)$$

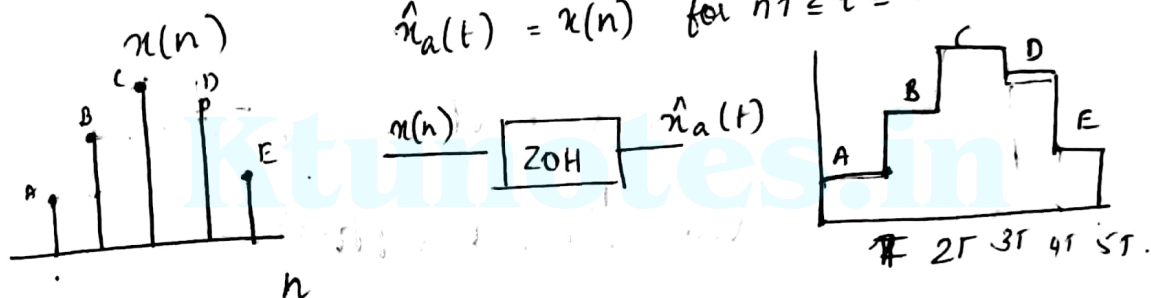
Impulse response of ZOH. $\xrightarrow{T.F.} \mathcal{L}\{h(t)\} = \frac{1}{s} - \frac{1}{s} e^{-Ts}$

$$H(s) = \frac{1 - e^{-Ts}}{s}$$

A ZOH filter is a more practical way to generate a sampled signal.



sample signal at a given time and holds value until next sampling instant.



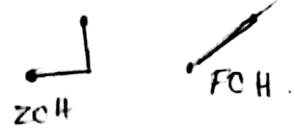
DT. signal $\hat{x}_a(t) = x(n)$ for $nT \leq t \leq (n+1)T$, $n=0,1,2,\dots$

Analog signal

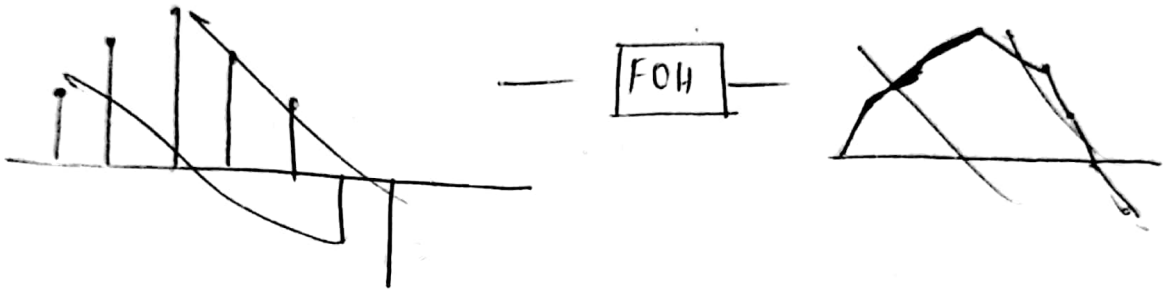
$\hat{x}_a(t) = x(0) \quad 0 \leq t \leq T$
 $= x(1) \quad T \leq t \leq 2T$
 $= x(2) \quad 2T \leq t \leq 3T$

The output of ZOH consists of higher order harmonics. To remove these harmonics the output of ZOH is applied to a low pass filter. This filter is called as smoothing filter b'cos it smooth the corners of the step approximations generated by ZOH.

First order hold



Adjacent samples are joined by straight lines



For the first order hold, the output $x_a(t)$ during interval $nT \leq t \leq (n+1)T$ is

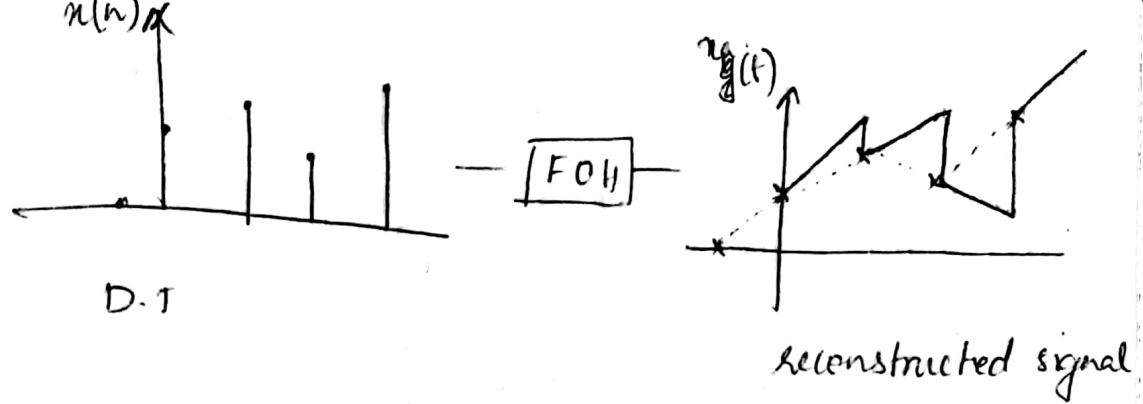
approximated as linear interpolation of previous two values

~~$$y(kT+1) = a_1 + a_0 \text{ where } a_0 = x(kT)$$~~

$$x_a(t) = x_a(nT) + \frac{x_a(nT) - x_a((n-1)T)}{T} (t - nT)$$

for $nT \leq t \leq (n+1)T$.

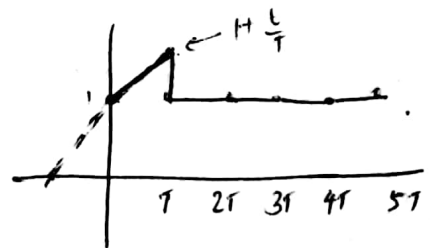
$n = 1, 2, 3, \dots$



First Order hold needs atleast two values to start construction of continuous signal, where as the ZOH needs one.

Impulse response of FOH is given by

$$h(t) = \begin{cases} 1 + \frac{t}{T} & \text{for } 0 \leq t \leq T \\ 1 - \frac{t}{T} & \text{for } T \leq t \leq 2T \\ 0 & \text{otherwise} \end{cases}$$



Impulse response and Convolution sum



$$y(n) = T[x(n)]$$

impulse response.

$$h(n) = T[\delta(n)]$$

Any signal $x(n)$ can be represented as

$$x(n) = \sum_{k=-\infty}^{\infty} x(k) \delta(n-k)$$

$$y(n) = T \left[\sum_{k=-\infty}^{\infty} x(k) \delta(n-k) \right]$$

$$= \sum_{k=-\infty}^{\infty} x(k) T[\delta(n-k)]$$

$\neq$

$$\delta(n) = 1 \quad n=0$$

$$\delta(n-k) = 1 \quad n=k$$

$$y(n) =$$

o/p to delayed impulse i/p

$$h(n, k) = T[\delta(n - k)]$$

For time invariant s/m

$$h(n, k) = h(n - k) = T[\delta(n - k)]$$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k) h(n - k)$$

$$y(n) = x(n) * h(n) \quad \text{Convolution sum}$$

Properties

The Convolution sum of two sequences can be found by using the following steps.

Step 1: Choose an initial value of n .

if $x(n)$ start at $n=n_1$, and $h(n)$ starts at $n=n_2$, then $n=n_1+n_2$ will be a good choice.

Step 2: Express both sequences in terms of the index k .

Step 3: Fold $h(k)$ about $k=0$ to obtain $h(-k)$ & shift by n to the right if n is positive and left if n is negative to obtain $h(n-k)$.

Step 4: Multiply the two sequences $x(k)$ and $h(n-k)$ element by element and sum the products to get $y(n)$.

Step 5: Increment the index n , shift the sequence $h(n-k)$ to right by one sample and do step 4.

Step 6: Repeat step 5 until the sum of products is zero for all remaining values of n .

→ Determine convolution sum of two sequence.

$$x(n) = \overset{-1}{1}, \overset{0}{4}, \overset{1}{3}, \overset{2}{2} \quad h(n) = \overset{0}{1}, \overset{1}{3}, \overset{2}{2}, \overset{3}{1}$$

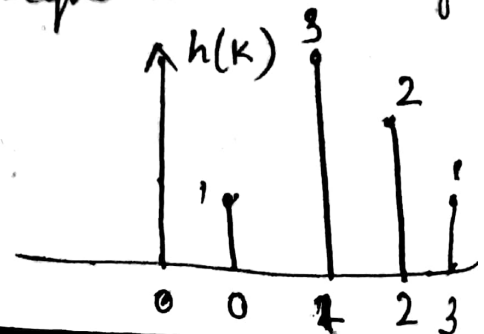
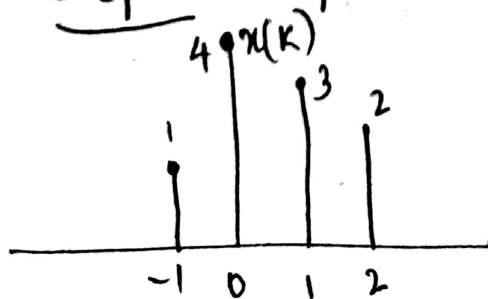
A: Step 1: Choose initial value of n .
 $n_1 = -1$ $n_2 = 0$

$$n(n) \quad L=4 \quad n = n_1 + n_2 = -1 + 0 = -1.$$

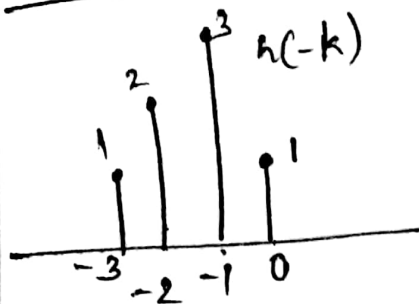
$$h(n) \quad M=4$$

No. of elements in o/p $y(n) \rightarrow L+M-1 = 4+4-1 = 7$

Step 2: Express both sequence in terms of k



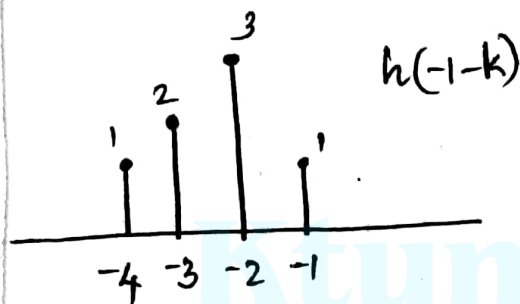
Step 3: Find $h(-k)$, then $h(n-k)$



$$\begin{array}{ll} h(0)=1 & h(-1)=3 \\ h(1)=3 & h(-2)=2 \\ h(2)=2 & h(-3)=1 \\ h(3)=1 & \end{array}$$

$$h(n-k) = h(-1-k)$$

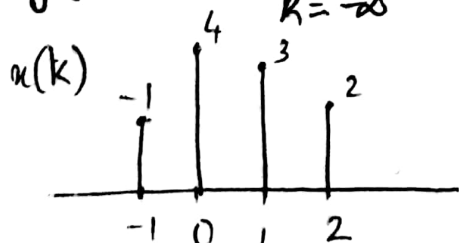
$n-k=0 \quad -1=k \quad k=-1$



$$\begin{array}{l} 1 \quad h(-1-0) = h(-1) = 3 \\ 1 \quad h(-1-1) = h(-2) = 2 \\ 2 \quad h(-1-2) = h(-3) = 1 \end{array}$$

Step 4: Multiply $x(k)$ and $h(-1-k)$ to find $y(n)$

$$y(-1) = \sum_{k=-\infty}^{\infty} x(k) h(-1-k)$$



$$n=-2 \quad y(-2)=0$$

$$n=-1 \quad y(-1) = -1 \times 1 = -1$$

$$n=0$$

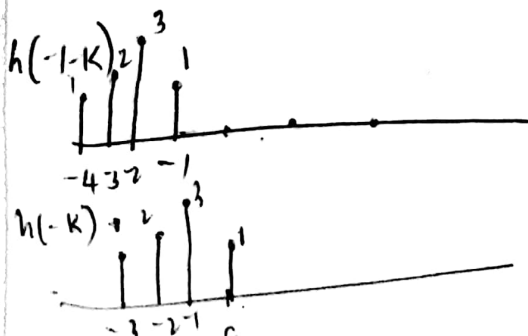
$$y(0) = 3 \times -1 + 4 \times 1 = -3 + 4 = 1$$

$$n=1$$

$$y(1) = 1 \times 3 + 3 \times 4 + 2 \times -1$$

$$n=2$$

$$y(2) = 2 \times 1 + 3 \times 3 + 4 \times 2 + -1 \times 1$$



Discrete - time convolution - Tabular method

$$1. x_1[n] = \{1, 2, -2\}$$

$\uparrow$
 $n=0$

$$x_2[n] = \{2, 0, 1\}$$

$\uparrow$
 $n=0$

$$y[n] = x_1[n] * x_2[n]$$

$x_2[n] \backslash x_1[n]$	1	2	-2
2	2	4	-4
0	0	0	0
-1	-1	-2	-2

$\uparrow$
 $n=0$

$$y[n] = \{2, 4, -3, 2, -2\}$$

$\uparrow$

• Multiply $x_2[n]$ with all elements of $x_1[n]$

Take elements diagonally by adding

To find value at $n=0$, take $x_1[0]$ and $x_2[0]$. Draw hori. & vert. line.

intersecting value is point and added value diagonally is marked.

$$2. \quad x_1[n] = \{-1, 2, 0, 1\}$$

$$x_2[n] = \{3, 1, 0, -1\}$$

$$y[n] = x_1[n] * x_2[n]$$

$x_1[n] \backslash x_2[n]$	-1	2	0	1
3	-3	6	0	3
1	-1	2	0	1
0	0	0	0	0
-1	1	-2	0	-1

$$y[n] = \{-3, +5, 2, 4, -1, 0, -1\}$$

$$3. \quad x_1[n] = \{1, -1, 1\}$$

$$x_2[n] = \{2, -1, 1\}$$

Find $y[n]$

I Obtain the convolution of the following sequences

1) $x(n) = u(n) - u(n-7)$

$h(n) = u(n-1) - u(n-4)$

2) using matrix method,

$x(n) = \{1, 4, 3, 1\}$ $h(n) = \{1, 2, 3, 2\}$

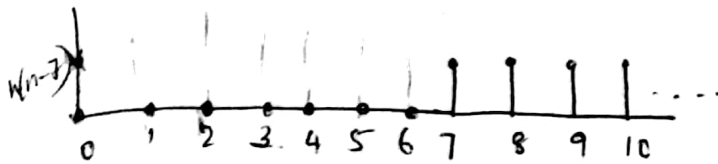
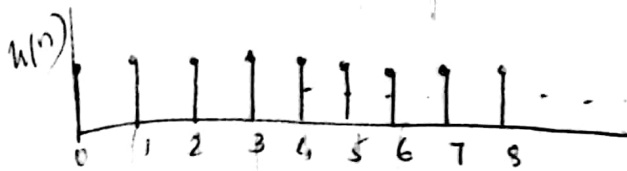
II → a. The impulse response of a s/m is given by $h(n) = [2, 3, 1]$. Find the response of the s/m when it is excited by the input $x(n) = u(n) - u(n-5)$.

→ b. Find the convolution sum given by

$x(n) = 2\delta(n+1) - \delta(n) + \delta(n-1) + 3\delta(n-2)$

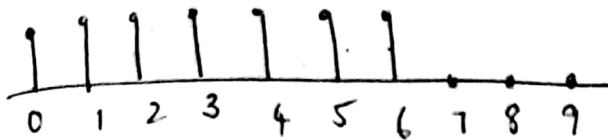
III Using graphical method, find convolution of $x[n] = \{1, 3, 3, 2\}$ and $h(n) = u(n) - u(n-4)$

$$x(n) = u(n) - u(n-7)$$



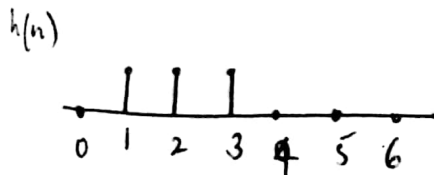
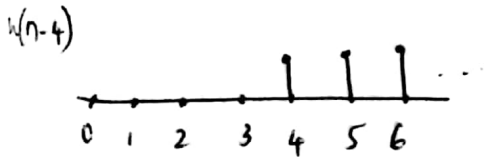
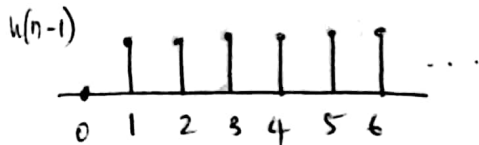
$$n-7=0 \\ n=7$$

$$x(n) =$$



$$x(n) = \{1, 1, 1, 1, 1, 1, 1\}$$

$$h(n) = u(n-1) - u(n-4)$$



$$h(n) = \{1, 1, 1\}$$

$$L = 7$$

$$M = 3$$

$$y(n) \rightarrow L+M-1 = 7+3-1 = 9$$

$$\text{starting} \rightarrow 1+0 = 1$$

$$\text{ending} \rightarrow 6+3 = 9$$

$x(n)$ $h(n)$							
	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1

$$y(n) = \{1, 2, 3, 3, 3, 3, 3, 2, 1\}$$

$$2) \quad x(n) = \{1, 4, 3, 1\} \quad h(n) = \{1, 2, 3, 2\}$$

$\begin{matrix} -3 & -2 & -1 & 0 \\ \uparrow & & & \end{matrix}$

$x(n)$ $h(n)$		1	4	3	1
1	1	4	3	(-1)	
2	2	8	6	2	
3	3	12	9	3	
2	2	8	6	2	

$$L = 4$$

$$M = 4$$

$$y(n) = 4 + 4 - 1 = 7$$

$$\text{starting} \rightarrow -3 + 0 = -3$$

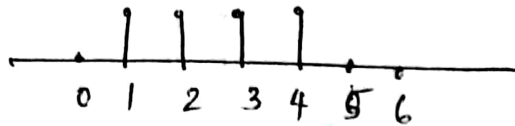
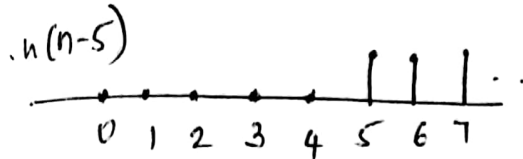
$$\text{ending} \rightarrow 0 + 3 = 3$$

$$y(n) = \{1, 6, 14, 21, 19, 9, 2\}$$

$\begin{matrix} -3 & -2 & -1 & 0 & 1 & 2 & 3 \\ & & & \uparrow & & & \end{matrix}$

$$1. a. \quad h(n) = \{2, 3, 1\}$$

$$x(n) = u(n-1) - u(n-5)$$



$$x(n) = \{1, 1, 1, 1\}$$

$$x[n] = \{0, 1, 1, 1, 1\}$$

$x(n)$ $h(n)$	1	1	1	1
2	2	2	2	2
3	3	3	3	3
1	1	1	1	1

$$y(n) = \{2, 5, 6, 6, 4, 1\}$$

$$L = 4$$

$$M = 3$$

$$y(n) \rightarrow 4 + 3 - 1 = 6$$

$$\text{start} \rightarrow 1 + 0 = 1$$

$$\text{end} \rightarrow 4 + 2 = 6$$

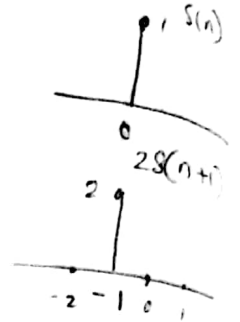
	0	1	1	1
2	0	2	2	2
3	0	3	3	3
1	0	1	1	1

$$y[n] = \{0, 2, 5, 6, 6, 4, 1\}$$

II b. $h(n) = \{2, 3, 1\}$

$$x(n) = 2s(n+1) - s(n) + s(n-1) + 3s(n-2)$$

$$= \{2, -1, 1, 3\}$$



$x(n) \backslash h(n)$	2	-1	1	3
2	4	-2	2	6
3	6	-3	3	9
1	2	-1	1	3

$L+M-1 = 4+3-1=6$
 start $\rightarrow -1+0 = -1$

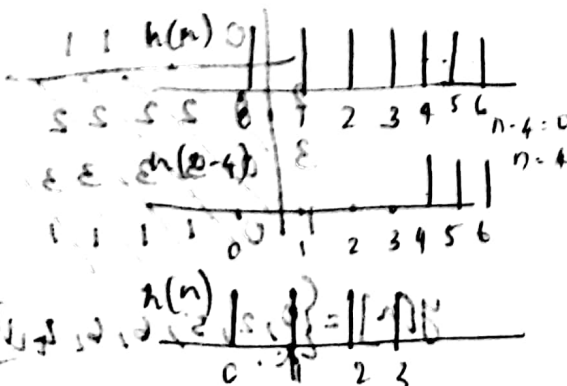
$y(n) = \{4, 4, 1, 8, 10, 3\}$

III. $x(n) = \{1, 3, 3, 2\}$

$L = 4$

$h(n) = u(n) - u(n-4)$

$M = 4$



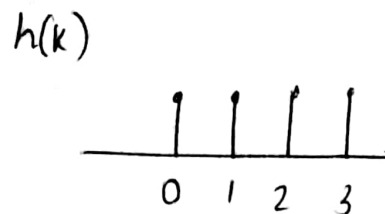
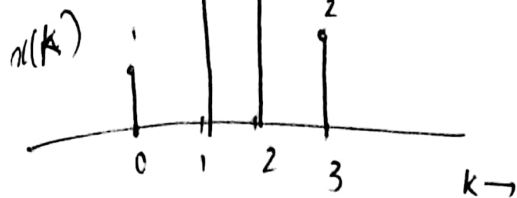
$y(n) \rightarrow 4+4-1=7$

starting $\rightarrow 0+0=0$

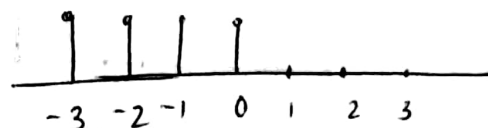
ending $\rightarrow 3+3=6$

$y(0)$ to $y(6)$.

By graphical method.

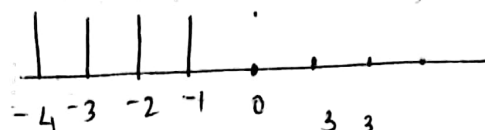


$h(-k)$



$h(n-k)$

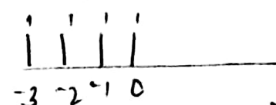
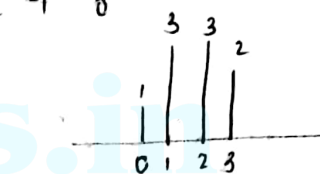
$n = -1$



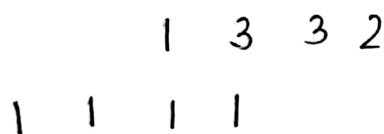
$n = 0$



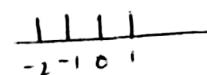
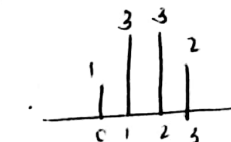
$$y(0) = 1$$



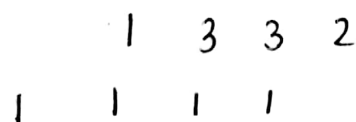
$n = 1$



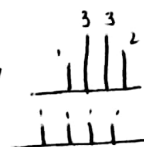
$$y(1) = 1 + 3 = 4$$



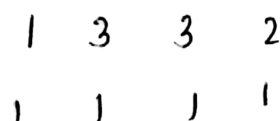
$n = 2$



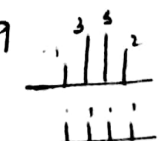
$$y(2) = 1 + 3 + 3 = 7$$



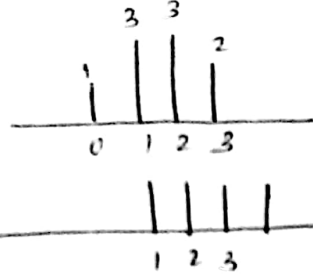
$n = 3$



$$y(3) = 1 + 3 + 3 + 2 = 9$$

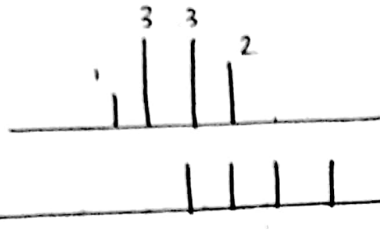


$$n = 4$$



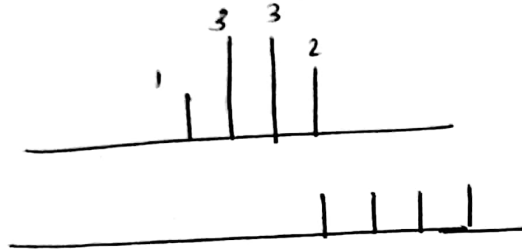
$$y(4) = 3 + 3 + 2 = 8$$

$$n = 5$$



$$y(5) = 3 + 2 = 5$$

$$n = 6$$



$$y(6) = 2$$

$$y(n) = \{1, 4, 7, 9, 8, 5, 2\}$$

$$\begin{array}{c|cccc} & 1 & 3 & 3 & 2 \\ 1 & & 1 & 3 & 3 & 2 \\ 1 & & & 3 & 3 & 2 \\ 1 & & & & 3 & 3 & 2 \\ 1 & & & & & 3 & 3 & 2 \\ 1 & & & & & & 3 & 3 & 2 \\ \hline \end{array}$$

Example 3.15: Determine the convolution sum of two sequences

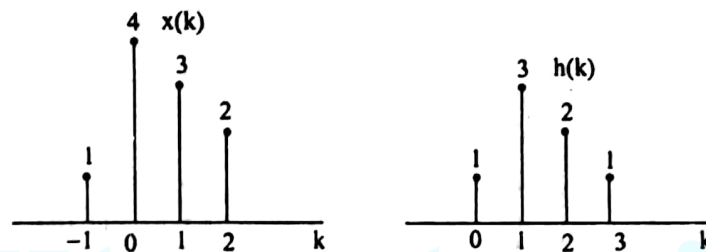
$$x(n) = \{1, 4, 3, 2\}; h(n) = \{1, 3, 2, 1\}$$

↑

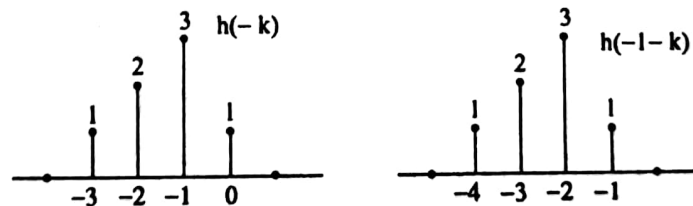
Solution

Step 1: The sequence $x(n)$ starts at $n_1 = -1$ and $h(n)$ starts at $n_2 = 0$. Therefore the starting time for evaluating the output sequence $y(n)$ is $n = n_1 + n_2 = -1 + 0 = -1$.

Step 2: Express both sequences in terms the index k .



Step 3: Fold $h(k)$ about $k = 0$, to obtain $h(-k)$



As starting time to evaluate $y(n)$ is -1 , shift $h(k)$ by one unit to left to obtain $h(-1-k)$.

$$y(-1) = \sum_{k=-\infty}^{\infty} x(k)h(-1-k)$$

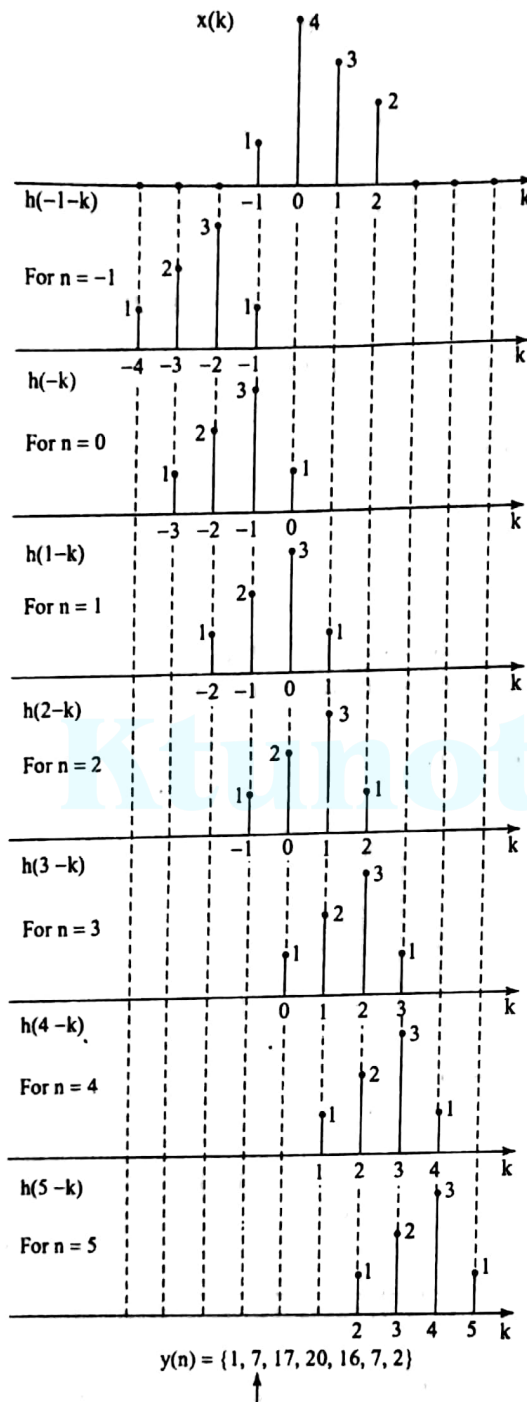


Fig. 3.9

Multiply the two sequences $x(k)$ and $h(-1-k)$ element by element and sum the products.

$$y(-1) = \sum_{k=-\infty}^{\infty} x(k)h(-1-k)$$

$$= (0)(1) + (0)(2) + 0(3) + (1)(1) + 4(0) + 3(0) + 2(0) = 1$$

Increment the index by 1, i.e., n to zero, shift the sequence to right to obtain $h(-k)$, and multiply the two sequences $x(k)$ and $h(-k)$ element by element and sum the products

$$y(0) = \sum_{k=-\infty}^{\infty} x(k)h(-k)$$

$$= (0)(1) + (0)(2) + 1(3) + 4(1) + 3(0) + 2(0) = 7$$

Similarly,

$$y(1) = \sum_{k=-\infty}^{\infty} x(k)h(1-k)$$

$$= 0(1) + 1(2) + 4(3) + 3(1) + 2(0) = 17$$

$$y(2) = \sum_{k=-\infty}^{\infty} x(k)h(2-k)$$

$$= 1(1) + 4(2) + 3(3) + 2(1) = 20$$

$$y(3) = \sum_{k=-\infty}^{\infty} x(k)h(3-k)$$

$$= 1(0) + 4(1) + 3(2) + 2(3) + 0(1) = 16$$

$$y(4) = \sum_{k=-\infty}^{\infty} x(k)h(4-k)$$

$$= 3(1) + 2(2) = 7$$

$$y(5) = \sum_{k=-\infty}^{\infty} x(k)h(5-k) = 2(1) = 2$$

For remaining values of n the value $y(n) = 0$. Therefore we stop at $n = 5$.

Example 3.16: Determine the output response $y(n]$ if

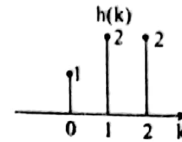
$$x(n) = \{1, 2, 3, 2\}; \quad h(n) = \{1, 2, 2\}$$

Solution Given

$$x(n) = \{1, 2, 3, 2\}; \quad h(n) = \{1, 2, 2\}$$

both the sequences starts at $n = 0$.

Therefore the output $y(n)$ also starts at $n = 0$.



We know

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

$$\begin{aligned} y(0) &= \sum_{k=-\infty}^{\infty} x(k)h(-k) \\ &= 1 \cdot 1 = 1 \end{aligned}$$

$$\begin{aligned} y(1) &= \sum_{k=-\infty}^{\infty} x(k)h(1-k) \\ &= 1 \cdot 2 + 2 \cdot 1 = 4 \end{aligned}$$

$$\begin{aligned} y(2) &= \sum_{k=-\infty}^{\infty} x(k)h(2-k) \\ &= 1 \cdot 2 + 2 \cdot 2 + 3 \cdot 1 = 9 \end{aligned}$$

$$\begin{aligned} y(3) &= \sum_{k=-\infty}^{\infty} x(k)h(3-k) \\ &= 2 \cdot 2 + 3 \cdot 2 + 2 \cdot 1 = 12 \end{aligned}$$

$$\begin{aligned} y(4) &= \sum_{k=-\infty}^{\infty} x(k)h(4-k) \\ &= 3 \cdot 2 + 2 \cdot 2 + 0 \cdot 1 = 10 \end{aligned}$$

$$\begin{aligned} y(5) &= \sum_{k=-\infty}^{\infty} x(k)h(5-k) \\ &= 2 \cdot 2 = 4 \end{aligned}$$

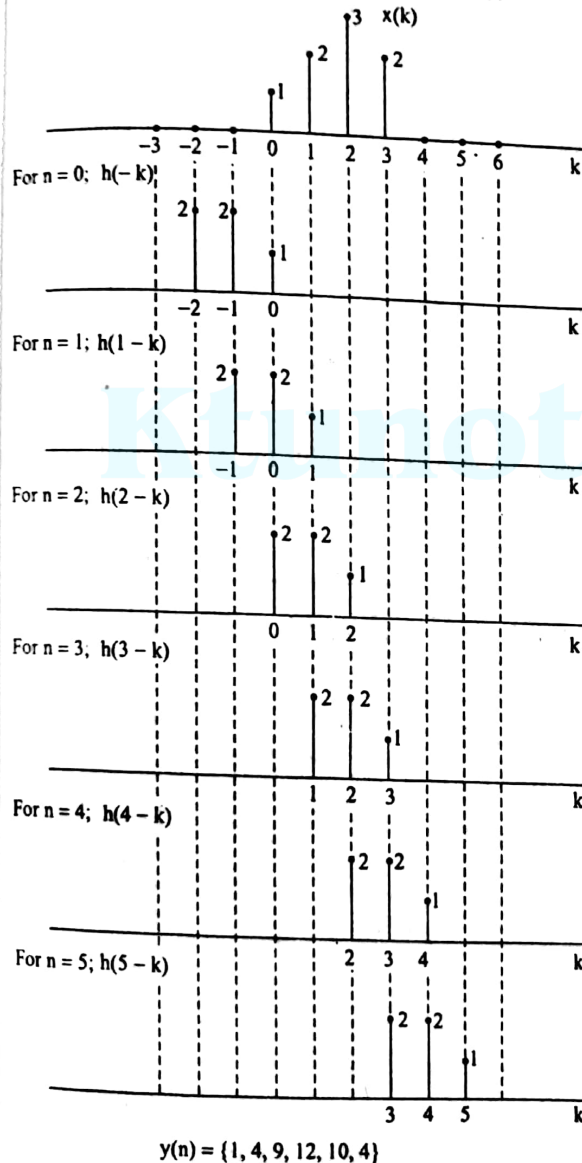


Fig. 3.10

The convolution of sequences $x_1(n)$ and $x_2(n)$ can also be found using following methods.

Z-Transform

ZT is equivalent to laplace transform.

$$CT \rightarrow LT$$

$$DT \rightarrow ZT$$

ZT is a method for the analysis of discrete time systems in frequency domain.

$$x[n] \xrightarrow[\text{IZT}]{\text{ZT}} X[z] = \sum_{n=-\infty}^{\infty} x[n] \cdot z^{-n}$$

$z \rightarrow$ Complex variable

$$z = r e^{j\omega}$$

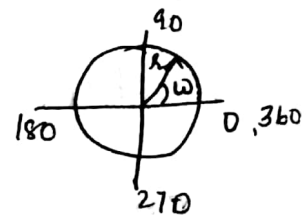
$$r = |z|$$

$\omega \rightarrow$ complex argument.

$$X[z] = \sum_{n=0}^{\infty} x[n] \cdot z^{-n} \rightarrow \text{unidirectional Z.T.}$$

→ Find the z-transform and ROC

for the signal $x(n) = a^n u(n)$.



$$u(n) = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

$$A: X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} a^n u(n) z^{-n} = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n$$

$$= 1 + (az^{-1}) + (az^{-1})^2 + (az^{-1})^3 + \dots \infty$$

Common ratio, C.R = az^{-1}

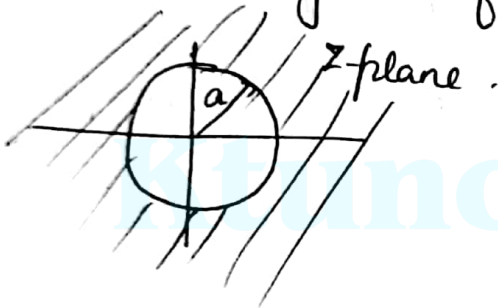
This is a geometric series of infinite length.

$$X(z) = \frac{1}{1 - (az^{-1})} \quad |az^{-1}| < 1$$

$$= \frac{z}{z - a} \quad \left| \frac{a}{z} \right| < 1$$

$$a < z \Rightarrow \underbrace{z > a}_{\text{Roc}}$$

ROC $\rightarrow$ Region of convergence.



Z.T exist only ^{within} at the ROC, i.e., outside of circle of radius a .

ROC of Z.T is a unit circle.

$\rightarrow$ Determine the Z-transform of the signal $x(n) = -b^n u(-n-1)$. Find ROC.

$$n: \quad u(-n-1) = 1 \quad n \leq -1$$

$$= 0 \quad n \geq -1$$

$$X(z) = \sum_{n=-\infty}^{\infty} -b^n u(-n-1) z^{-n}$$

$$= \sum_{n=-\infty}^{-1} -b^n z^{-n}$$

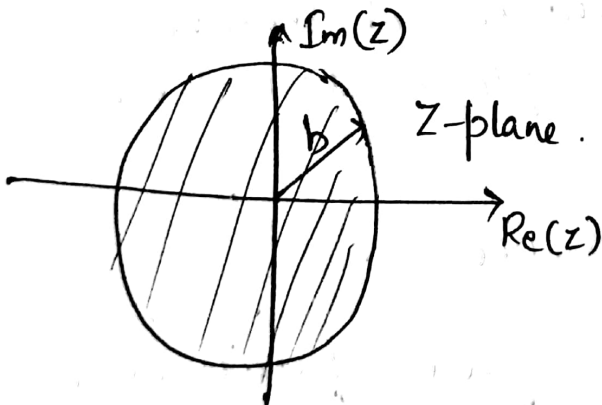
$$= - \sum_{n=-\infty}^{-1} (bz^{-1})^n$$

$$= - \sum_{n=1}^{\infty} (b^{-1}z)^n$$

$$= - \left[b^{-1}z + (b^{-1}z)^2 + (b^{-1}z)^3 + \dots \infty \right]$$

$$= - \frac{b^{-1}z}{1 - b^{-1}z} = \frac{z}{z-b} \quad |b^{-1}z| < 1$$

$$\underbrace{|z| < b}_{\text{Roc}}$$



Z-Transform and ROC of finite duration

Sequence

1. Right hand sequence.

$$x(n) = 0 \quad n < n_0, \quad n_0 \text{ is finite.}$$

If $n_0 \geq 0 \rightarrow$ causal sequence.

ROC is entire z-plane except at $z=0$.

2. Left hand sequence

$$x(n) = 0 \quad n \geq n_0.$$

$n_0 \leq 0 \rightarrow$ anticausal sequence.

ROC is entire z-plane except at $z=\infty$.

3. Two sided sequence.

A signal has finite duration on both left and right sides. is known as two-sided sequence.

Roc is entire z -plane except $z=0$ & $z=\infty$.

→ Find z -transform and Roc of the causal sequence.

$$x(n) = \{2, -1, 3, 2, 0, 1\}$$

↑

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$= \sum_{n=0}^5 x(n) z^{-n}$$

$$= x(0) + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3} + x(4)z^{-4} + x(5)z^{-5}$$

$$= 2 + -1z^{-1} + 3z^{-2} + 2z^{-3} + z^{-5}$$

Roc → entire z -plane except $z=0$

→ Find the Z-T and Roc of the anticausal system.

$$x(n) = \overset{n=-4}{\{3, 2, -1, -4, \overset{0}{-1}\}} \quad \uparrow$$

$$\begin{aligned} A: \quad X(z) &= \sum_{n=-4}^0 x(n) z^{-n} \\ &= x(-4)z^{+4} + x(-3)z^{+3} + x(-2)z^{+2} + x(-1)z^{+1} + x(0) \\ &= 3z^{+4} + 2z^{+3} - z^{+2} - 4z^{+1} - 1 \end{aligned}$$

$X(z)$ converges for all values of z except at $z = \infty$.

→ Find the ZT of the sequence.

$$x(n) = \overset{-4}{\overset{-3}{\overset{-2}{\overset{-1}{\overset{0}{\overset{1}{\overset{2}{\overset{3}{\overset{4}{\uparrow}}}}}}}} \{1, 2, 0, -4, 3, 2, 1, 6, 5\}$$

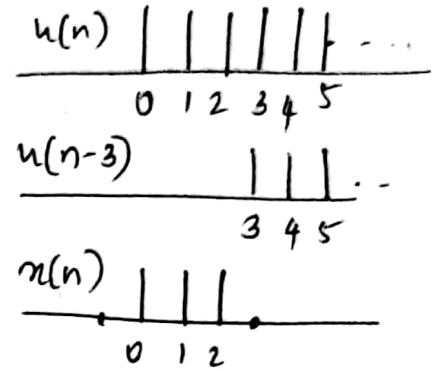
$$\begin{aligned} A: \quad X(z) &= \sum_{n=-4}^4 x(n) z^{-n} \\ &= z^4 + 2z^3 - 4z + 3 + 2z^{-1} + z^{-2} + 6z^{-3} + 5z^{-4} \end{aligned}$$

$X(z)$ converges for all values of z except $z=0$ & $z=\infty$

→ Find the ZT of the following sequence

i) $x(n) = u(n) - u(n-3)$

$$X(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$$
$$= 1 + z^{-1} + z^{-2}$$



ROC → Entire z -plane except $z = 0$.

ii) $x(n) = \{1, 2, 3, 2\}$

iii) $x(n) = \{1, 2, -1, 2, 3\}$

↑

Properties of Region of Convergence

→ Find the ZT of the signal

$$x[n] = (\sin(\omega_0 n)) u[n] \text{ and find ROC.}$$

$$A: X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} \sin \omega_0 n u[n] z^{-n} \quad u[n] = 1 \text{ for } n \geq 0$$

$$= \sum_{n=0}^{\infty} (\sin \omega_0 n) z^{-n} \quad \sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$= \sum_{n=0}^{\infty} \frac{e^{j\omega_0 n} - e^{-j\omega_0 n}}{2j} z^{-n}$$

$$= \frac{1}{2j} \left[\sum_{n=0}^{\infty} e^{j\omega_0 n} z^{-n} - \sum_{n=0}^{\infty} e^{-j\omega_0 n} z^{-n} \right]$$

$$= \frac{1}{2j} \left[\sum_{n=0}^{\infty} (e^{j\omega_0} z^{-1})^n - \sum_{n=0}^{\infty} (e^{-j\omega_0} z^{-1})^n \right]$$

$$= \frac{1}{2j} \left[\frac{1}{1 - e^{j\omega_0} z^{-1}} - \frac{1}{1 - e^{-j\omega_0} z^{-1}} \right]$$

$$\begin{aligned} |e^{j\omega_0} z^{-1}| &< 1 & \& \\ |e^{-j\omega_0} z^{-1}| &< 1 & \& \end{aligned}$$

$$= \frac{1}{2j} \left[\frac{z}{z - e^{j\omega_0}} - \frac{z}{z - e^{-j\omega_0}} \right] \quad \begin{array}{l} |e^{j\omega_0}| \neq z \quad |e^{-j\omega_0}| \neq z \\ \cancel{z \neq} \quad |z| > 1 \end{array}$$

$$= \frac{1}{2j} \left[\frac{z(z - e^{-j\omega_0}) - z(z - e^{j\omega_0})}{(z - e^{j\omega_0})(z - e^{-j\omega_0})} \right]$$

$$= \frac{1}{2j} \left[\frac{z^2 - z e^{-j\omega_0} - \cancel{z^2} + z e^{j\omega_0}}{z^2 - z e^{-j\omega_0} - z e^{j\omega_0} + 1} \right]$$

$$= \frac{1}{2j} \left[\frac{z(e^{j\omega_0} - e^{-j\omega_0})}{z^2 - z(e^{j\omega_0} + e^{-j\omega_0}) + 1} \right]$$

$$x(z) = \frac{z \sin \omega_0}{z^2 - 2z \cos \omega_0 + 1}$$

$$\text{ROC} : |z| > 1$$

$$= \frac{(\sin \omega_0) z^{-1}}{1 - 2(\cos \omega_0) z^{-1} + z^{-2}}$$

→ Determine the z-transform, ROC and pole-zero locations of $x(z)$ for

$$x(n) = \left(\frac{2}{3}\right)^n u(n) + \left(-\frac{1}{2}\right)^n u(n).$$

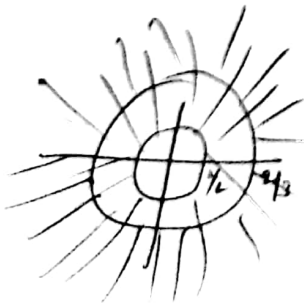
$$\begin{aligned} \text{A: } X(z) &= \sum_{n=-\infty}^{\infty} x(n) z^{-n} \\ &= \sum_{n=-\infty}^{\infty} \left[\left(\frac{2}{3}\right)^n u(n) + \left(-\frac{1}{2}\right)^n u(n) \right] z^{-n} \end{aligned}$$

$$= \sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n z^{-n} + \sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{2}{3} z^{-1}\right)^n + \sum_{n=0}^{\infty} \left(-\frac{1}{2} z^{-1}\right)^n$$

$$= \frac{1}{1 - \frac{2}{3} z^{-1}} + \frac{1}{1 + \frac{1}{2} z^{-1}}$$

$$= \frac{z}{z - \frac{2}{3}} + \frac{z}{z + \frac{1}{2}} \quad \begin{matrix} \left| \frac{2}{3} z^{-1} \right| < 1 \\ \left| -\frac{1}{2} z^{-1} \right| < 1 \end{matrix}$$



$$|z| > \frac{2}{3} \text{ \& } |z| > \frac{1}{2}$$

$$\text{ROC} \rightarrow |z| > \frac{2}{3}$$

$$X(z) = \frac{z(z + \frac{1}{2}) + z(z - \frac{2}{3})}{(z + \frac{1}{2})(z - \frac{2}{3})}$$

$$= \frac{z^2 + \frac{z}{2} + z^2 - \frac{2z}{3}}{(z + \frac{1}{2})(z - \frac{2}{3})}$$

$$= \frac{2z^2 - \frac{1}{6}z}{(z + \frac{1}{2})(z - \frac{2}{3})} = \frac{z(2z - \frac{1}{6})}{(z + \frac{1}{2})(z - \frac{2}{3})}$$

$$\frac{1}{2} - \frac{2}{3}$$

$$\frac{3-4}{2 \times 3}$$

$$\frac{-1}{2 \times 3}$$

$$-\frac{1}{6}$$

Zeros at $z = 0, \frac{1}{12}$

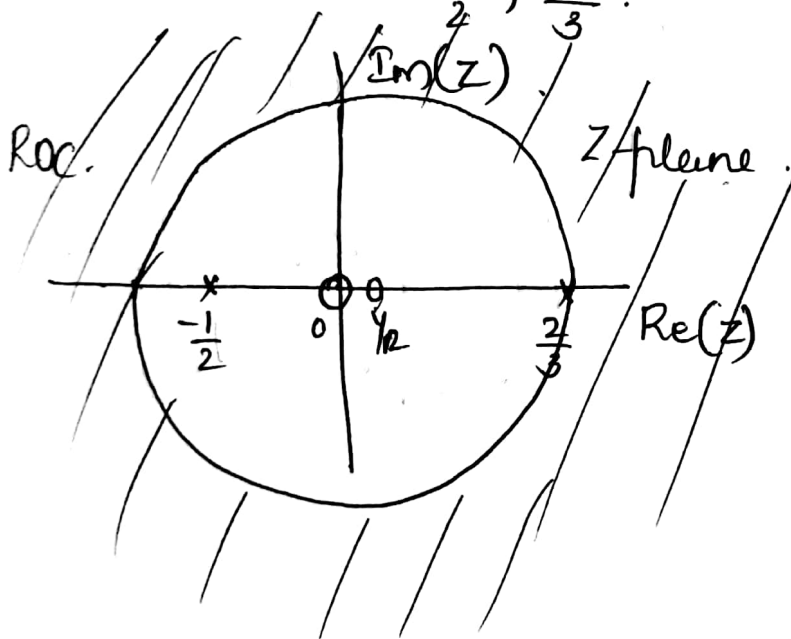
$$\text{Roc: } |z| > \frac{2}{3}$$

$$2z - \frac{1}{6} = 0$$

$$2z = \frac{1}{6}$$

$$z = \frac{1}{12}$$

Poles at $z = -\frac{1}{2}, \frac{2}{3}$

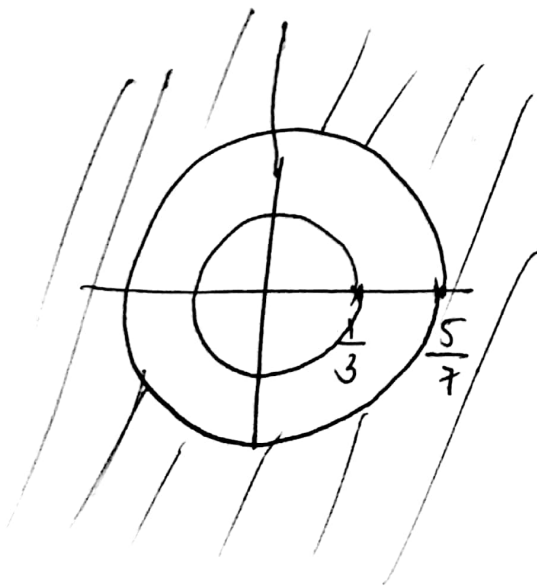


→ Find z_T & ROC

$$x[n] = 3 \left(\frac{5}{7}\right)^n u(n) + 2 \left(-\frac{1}{3}\right)^n u(n).$$

Also find pole-zero location.

$$\begin{aligned} A: X(z) &= \sum_{n=-\infty}^{\infty} \left[3 \left(\frac{5}{7}\right)^n u(n) + 2 \left(-\frac{1}{3}\right)^n u(n) \right] z^{-n} \\ &= \sum_{n=0}^{\infty} 3 \left(\frac{5}{7}\right)^n z^{-n} + \sum_{n=0}^{\infty} 2 \left(-\frac{1}{3}\right)^n z^{-n} \\ &= 3 \sum_{n=0}^{\infty} \left(\frac{5}{7} z^{-1}\right)^n + 2 \sum_{n=0}^{\infty} \left(-\frac{1}{3} z^{-1}\right)^n \\ &= 3 \frac{z}{z - \frac{5}{7}} + 2 \frac{z}{z + \frac{1}{3}} \end{aligned}$$



$$\left| \frac{5}{7} z^{-1} \right| < 1 \quad \& \quad \left| -\frac{1}{3} z^{-1} \right| < 1$$

$$\& \left| z \right| > \frac{5}{7} \quad \left| z \right| > \frac{1}{3}$$

0.7 0.3

$$ROC \rightarrow \left| z \right| > \frac{5}{7}$$

$$X(z) = 3 \frac{z}{z - 5/7} + \frac{2z}{z + 1/3}$$

$$= \frac{3z(z + 1/3) + 2z(z - 5/7)}{(z - 5/7)(z + 1/3)}$$

$$= \frac{z \left[3z + 1 + 2z - 10/7 \right]}{(z - 5/7)(z + 1/3)}$$

$$\begin{aligned} 1 - \frac{10}{7} \\ \frac{7-10}{7} = -\frac{3}{7} \end{aligned}$$

$$= \frac{z \left[5z - \frac{3}{7} \right]}{(z - 5/7)(z + 1/3)} ; \text{ROC} ; |z| > \frac{5}{7}$$

Zeros $\rightarrow z = 0, \frac{3}{35}$

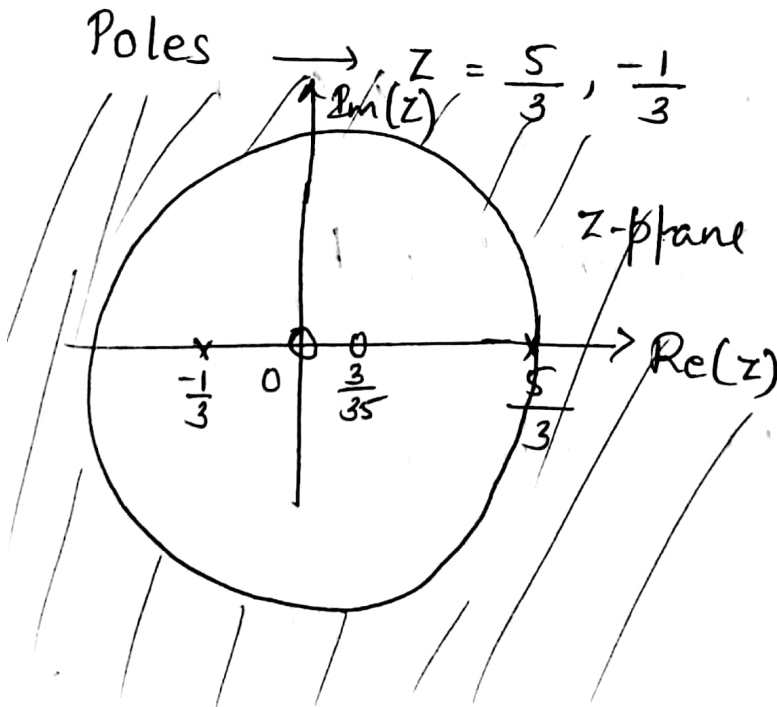
$$5z - \frac{3}{7} = 0$$

$$5z = \frac{3}{7}$$

$$z = \frac{3}{35}$$

Poles

$$z = \frac{5}{7}, -\frac{1}{3}$$



Properties of z-Transform:

2.

1. Linearity

States that the z-transform of a weighted sum of two signals is equal to the weighted sum of the individual z-transforms.

$$x_1(z) = Z\{x_1(n)\} \quad x_2(z) = Z\{x_2(n)\}$$

$$Z\{ax_1(n) + bx_2(n)\} = ax_1(z) + bx_2(z)$$

Proof:

$$Z\{ax_1(n) + bx_2(n)\} = \sum_{n=-\infty}^{\infty} [ax_1(n) + bx_2(n)]z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} ax_1(n)z^{-n} + b \sum_{n=-\infty}^{\infty} x_2(n)z^{-n}$$

$$= ax_1(z) + bx_2(z)$$

ROC $\rightarrow$ intersection of ROCs of individual z.T.

Time Shifting

If $x(z) = Z\{x(n)\}$ & initial conditions for $x(n)$ are zero, then.

$$Z\{x(n-m)\} = z^{-m} x(z)$$

Proof.

$$Z\{x(n-m)\} = \sum_{n=-\infty}^{\infty} x(n-m) z^{-n}$$

$$\text{Let } n-m=p \Rightarrow n=p+m.$$

$$Z\{x(n-m)\} = \sum_{p=-\infty}^{\infty} x(p) z^{-(p+m)}$$

$n = -\infty$
 $p = n-m$
 $= -\infty$

$$= z^{-m} \sum_{p=-\infty}^{\infty} x(p) z^{-p}$$

$$\boxed{Z\{x(n-m)\} = z^{-m} x(z)}$$

If $m > 0$, ROC $\rightarrow$ ^{same of $x(z)$} ~~entire z -plane~~ except $z=0$.

3. Time Reversal

If $x(z) = Z\{x(n)\}$, then $ROC \rightarrow R$

$$Z\{x(-n)\} = x(z^{-1}) \quad ROC \rightarrow R^{-1} \text{ or } \frac{1}{R}$$

Proof:

$$Z\{x(-n)\} = \sum_{n=-\infty}^{\infty} x(-n) z^{-n}$$

Let $-n = u \Rightarrow n = -u$
 $n = -\infty \rightarrow u = \infty$
 $n = \infty \rightarrow u = -\infty$

$$= \sum_{u=\infty}^{-\infty} x(u) z^u$$

$$= \sum_{u=-\infty}^{\infty} x(u) z^{+u}$$

$$= \sum_{u=-\infty}^{\infty} x(u) (z^{-1})^{-u}$$

$$\boxed{Z\{x(-n)\} = x(z^{-1})}$$

4. Time Scaling

$$x(n) \xleftrightarrow{Z.T} X(z) \quad ROC = R$$

$$x\left(\frac{n}{m}\right) \xleftrightarrow{\quad} X(z^m) \quad ROC = R^{1/m}$$

Proof:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$Z\left\{x\left(\frac{n}{m}\right)\right\} = \sum_{n=-\infty}^{\infty} x\left(\frac{n}{m}\right) z^{-n}$$

$$\text{Let } \frac{n}{m} = k \Rightarrow n = mk$$

$$n = -\infty \quad k = -\infty$$

$$n = \infty \quad k = \infty$$

$$Z\left\{x\left(\frac{n}{m}\right)\right\} = \sum_{k=-\infty}^{\infty} x(k) z^{-mk}$$

$$= \sum_{k=-\infty}^{\infty} x(k) (z^m)^{-k}$$

$$\boxed{Z\left\{x\left(\frac{n}{m}\right)\right\} = X(z^m)}$$

5. Multiplication by exponential sequence.

If $x(z) \longrightarrow Z\{x(n)\}$, then

$$Z\{a^n x(n)\} = x(a^{-1}z)$$

Proof:

$$\begin{aligned}
 Z\{a^n x(n)\} &= \sum_{n=-\infty}^{\infty} a^n x(n) z^{-n} \\
 &= \sum_{n=-\infty}^{\infty} x(n) (a z^{-1})^n \\
 &= \sum_{n=-\infty}^{\infty} x(n) (a^{-1} z)^{-n} \\
 &= X(a^{-1} z)
 \end{aligned}$$

ROC is $R_1 < |a^{-1} z| < R_2$.

$$|a| R_1 < |z| < |a| R_2$$

6. Convolution

$$Z\{x(n)\} = X(z) ; Z\{h(n)\} = H(z)$$

$$Z\{x(n) * h(n)\} = X(z) \cdot H(z)$$

Proof:

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k)$$

$$Z\{x(n) * h(n)\} = Z\{y(n)\} = Y(Z)$$

$$\begin{aligned} Y(Z) &= \sum_{n=-\infty}^{\infty} y(n) z^{-n} \\ &= \sum_{n=-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} x(k) h(n-k) \right] z^{-n} \end{aligned}$$

Interchanging the order of summation.

$$\begin{aligned} Y(Z) &= \sum_{k=-\infty}^{\infty} x(k) \sum_{n=-\infty}^{\infty} h(n-k) z^{-n} \\ &= \sum_{k=-\infty}^{\infty} x(k) z^{-k} \sum_{n=-\infty}^{\infty} h(n-k) z^{-(n-k)} \\ &= \cancel{X(Z)} \\ &= \sum_{k=-\infty}^{\infty} x(k) z^{-k} \sum_{m=-\infty}^{\infty} h(m) z^{-m} \end{aligned}$$

Let $n-k=m$
 $n=-\infty \quad m=-\infty$
 $n=\infty \quad m=\infty$

$\sum_{m=-\infty}^{\infty}$

$$\boxed{Y(Z) = X(Z) \cdot H(Z)}$$

Imp.
7.

Initial Value Theorem.

If $x_+(z) = Z\{x(n)\}$, then

$$x(0) = \lim_{z \rightarrow \infty} x_+(z)$$

Proof:

$$x_+(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$$

$$= x(0) + x(1)z^{-1} + x(2)z^{-2} + \dots$$

As $z \rightarrow \infty$, all terms vanish except $x(0)$.

$$\lim_{z \rightarrow \infty} x_+(z) = \underline{\underline{x(0)}}$$

8. Final Value Theorem., $\forall x(n)=0; n < 0$

$$x(\infty) = \lim_{z \rightarrow 1} (z-1) x_+(z)$$

(z-1)x(z) should have poles inside unit circle in z-plane.

Proof:

$$\cancel{Z\{x(n+1)\}} - \cancel{Z\{x(n)\}} = \sum_{n=0}^{\infty} (x(n+1) - x(n)) z^{-n}$$

$$Z \{x(n+1) - x(n)\} = \sum_{n=0}^{\infty} [x(n+1) - x(n)] Z^{-n}$$

linearity prop.

$$Z \{x(n+1)\} - Z \{x(n)\} = \sum_{n=0}^{\infty} [x(n+1) - x(n)] Z^{-n}$$

$$Z \cdot x(z) - x(0) = \sum_{n=0}^{\infty} [x(n+1) - x(n)] Z^{-n}$$

$$= (Z-1) x(z) - x(0) = \sum_{n=0}^{\infty} [x(n+1) - x(n)] Z^{-n}$$

Let $z \rightarrow 1$

$$\lim_{z \rightarrow 1} (z-1)x(z) - x(0) = z^{-1} [x(1) - x(0) + x(2) - x(1) + x(3) - x(2) + x(4) - x(3) + \dots + x(\infty+1) - x(\infty)]$$

$$\boxed{\lim_{z \rightarrow 1} (z-1)x(z) = x(\infty)}$$

Table 10.1 z-transform pairs.

	Sequence $x(n)$	z-Transform $X(z)$	ROC
1.	$\delta(n)$	1	All z
2.	$u(n)$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
3.	$-u(-n - 1)$	$\frac{1}{1 - z^{-1}}$	$ z < 1$
4.	$\delta(n - m)$	z^{-m}	All z except at $z = 0$ (if $m > 0$) All z except at $z = \infty$ (if $m < 0$)
5.	$a^n u(n)$	$\frac{1}{1 - az^{-1}}$	$ z > a $
6.	$-a^n u(-n - 1)$	$\frac{1}{1 - az^{-1}}$	$ z < a $
7.	$na^n u(n)$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z > a $
8.	$-na^n u(-n - 1)$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z < a $
9.	$[\cos \omega_0 n] u(n)$	$\frac{1 - (\cos \omega_0) z^{-1}}{1 - (2 \cos \omega_0) z^{-1} + z^{-2}}$	$ z > 1$
10.	$[\sin \omega_0 n] u(n)$	$\frac{(\sin \omega_0) z^{-1}}{1 - (2 \cos \omega_0) z^{-1} + z^{-2}}$	$ z > 1$
11.	$r^n \cos \omega_0 n u(n)$	$\frac{1 - (r \cos \omega_0) z^{-1}}{1 - 2r(\cos \omega_0) z^{-1} + r^2 z^{-2}}$	$ z > r $
12.	$r^n \sin \omega_0 n u(n)$	$\frac{r \sin \omega_0 z^{-1}}{1 - 2r \cos \omega_0 z^{-1} + r^2 z^{-2}}$	$ z > r $
13.	$\frac{1}{n}, n > 0$	$-\ln(1 - z^{-1})$	$1 < z $
14.	$n^k a^n, k < 0$	$- \left(-z \frac{d}{dt} \right)^n \frac{1}{1 - az^{-1}}$	$ z < a $
15.	$a^{ n }$ all n	$\frac{1 - a^2}{(1 - az)(1 - az^{-1})}$	$ a < z < \left \frac{1}{a} \right $

The Inverse z-Transform.

$$x(n) = z^{-1} [X(z)]$$

1. Long Division method.
2. Partial fraction method
3. Residue method.
4. Convolution method.

Long Division method.

→ Using long division, determine the inverse z-transform of

$$X(z) = \frac{1 + 2z^{-1}}{1 - 2z^{-1} + z^{-2}}$$

- if (a) $x(n)$ is causal.
 (b) is anti causal.

Causal $\rightarrow$ negative powers of z (z^{-1})

begin division with constant

→ positive powers of z (z')

begin division with highest power of x .

Anti causal $\rightarrow z^{-1} \rightarrow$ highest power of z^{-1}
 $z \rightarrow$ constant or lower power

$$\begin{array}{r}
 1 - 2z^{-1} + z^{-2} \overline{) 1 + 4z^{-1} + 7z^{-2} + 10z^{-3} + 13z^{-4} + 16z^{-5} + \dots} \\
 \underline{1 + 2z^{-1}} \phantom{+ 7z^{-2} + 10z^{-3} + 13z^{-4} + 16z^{-5} + \dots} \\
 1 - 2z^{-1} + z^{-2} \\
 \underline{+} \\
 4z^{-1} - z^{-2} \\
 \underline{4z^{-1} - 8z^{-2} + 4z^{-3}} \\
 7z^{-2} - 7z^{-3} \\
 \underline{7z^{-2} - 14z^{-3} + 7z^{-4}} \\
 10z^{-3} - 7z^{-4} \\
 \underline{10z^{-3} - 20z^{-4} + 10z^{-5}} \\
 13z^{-4} - 10z^{-5} \\
 \underline{13z^{-4} - 26z^{-5} + 13z^{-6}} \\
 16z^{-5} - 13z^{-6}
 \end{array}$$

$$X(z) = 1 + 4z^{-1} + 7z^{-2} + 10z^{-3} + 13z^{-4} + 16z^{-5} + \dots$$

$$x(n) = \{1, 4, 7, 10, 13, 16, \dots\}$$

(b) Anticausal $\uparrow$

$$\begin{array}{r}
 z^{-2} - 2z^{-1} + 1 \overline{) 2z + 5z^2 + 8z^3 + 11z^4 + 14z^5} \\
 \underline{2z^{-1} + 1} \\
 2z^{-1} - 4 + 2z \\
 \underline{+ 4} \\
 5 - 2z \\
 \underline{5 - 10z + 5z^2} \\
 8z - 5z^2 \\
 \underline{8z - 16z^2 + 8z^3} \\
 11z^2 - 8z^3 \\
 \underline{11z^2 - 22z^3 + 11z^4} \\
 14z^3 - 11z^4
 \end{array}$$

$$X(z) = 2z + 5z^2 + 8z^3 + 11z^4 + 14z^5 + \dots$$

$$x(n) = \{ \dots, 14, 11, 8, 5, 2, 0 \}$$

$\uparrow$

→ Using long division, determine ZF inv. ZT.

of $X(z) = \frac{z+1}{z^2-3z+2}$ if

(a) $x(n)$ is causal

(b) $x(n)$ is anticausal.

$$\begin{array}{r}
 z^{-1} + 4z^{-2} + 10z^{-3} + 22z^{-4} \\
 \hline
 z^2 - 3z + 2 \overline{) \begin{array}{l} z + 1 \\ z - 3 + 2z^{-1} \\ \hline 4 - 2z^{-1} \\ 4 - 12z^{-1} + 8z^{-2} \\ \hline 10z^{-1} - 8z^{-2} \\ 10z^{-1} - 40z^{-2} + 20z^{-3} \\ \hline 22z^{-2} - 20z^{-3} \\ 22z^{-2} - 66z^{-3} + 44z^{-4} \\ \hline 46z^{-3} - 44z^{-4} \end{array} }
 \end{array}$$

$$X(z) = z^{-1} + 4z^{-2} + 10z^{-3} + 22z^{-4} + \dots$$

$$x(n) = \{ 0, 1, 4, 10, 22, \dots \}$$

↑

(b) Anticausal

$$\begin{array}{r}
 \frac{1}{2} + \frac{5}{4}z + \frac{13}{8}z^2 + \dots \\
 2-3z+z^2 \overline{) 1+z} \\
 \underline{1-\frac{3}{2}z+\frac{1}{2}z^2} \\
 \frac{5}{2}z - \frac{1}{2}z^2 \\
 \frac{5}{2}z - \frac{15}{4}z^2 + \frac{5}{4}z^3 \\
 \underline{+\frac{5}{4}z^2 - \frac{5}{4}z^3} \\
 \frac{13}{4}z^2 - \frac{39}{8}z^3 + \frac{13}{8}z^4 \\
 \underline{+\frac{13}{8}z^3 - \frac{13}{8}z^4} \\
 \frac{29}{8}z^3 - \frac{13}{8}z^4
 \end{array}$$

$$2 \times \frac{5}{4} = \frac{5}{2}$$

$$\frac{15}{4} - \frac{1}{2}$$

$$\frac{15-2}{4}$$

$$\frac{13}{4}$$

$$2 \times \frac{13}{8} = \frac{13}{4}$$

$$\frac{39}{8} - \frac{5}{4}$$

$$\frac{39-10}{8}$$

$$\frac{29}{8}$$

$$x(z) = \frac{1}{2} + \frac{5}{4}z + \frac{13}{8}z^2 + \dots$$

$$x(n) = \left\{ \dots, \frac{13}{8}, \frac{5}{4}, \frac{1}{2} \right\}$$



Partial Fraction Expansion Method.

$$X(z) = \frac{N(z)}{D(z)} = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}.$$

Multiply $N(z)$ & $D(z)$ by z^N to eliminate negative powers of z .

$$X(z) = \frac{b_0 z^N + b_1 z^{N-1} + \dots + b_M z^{N-M}}{z^N + a_1 z^{N-1} + \dots + a_N}$$

Divide by z

$$\begin{aligned} \frac{X(z)}{z} &= \frac{b_0 z^{N-1} + b_1 z^{N-2} + \dots + b_M z^{N-M-1}}{z^N + a_1 z^{N-1} + \dots + a_N} \\ &= \frac{b_0 z^{N-1} + b_1 z^{N-2} + \dots + b_M z^{N-M-1}}{(z-p_1)(z-p_2) \dots (z-p_N)} \end{aligned}$$

For distinct poles .

$$\frac{X(z)}{z} = \frac{C_1}{z-p_1} + \frac{C_2}{z-p_2} + \dots + \frac{C_N}{z-p_N}$$

$$C_k = (z-p_k) \frac{X(z)}{z} \Big|_{z=p_k} \quad k=1, 2, \dots, N$$

$(z-p_k)^l \rightarrow$ multiple poles .

eg: $\frac{X(z)}{z} = \frac{1}{(z-p_1)(z-p_2)^2}$

$$\frac{X(z)}{z} = \frac{C_1}{z-p_1} + \frac{C_2}{z-p_2} + \frac{C_3}{(z-p_2)^2}$$

$$C_1 = (z-p_1) \frac{X(z)}{z} \Big|_{z=p_1}$$

$$C_2 = \frac{d}{dz} \left[(z-p_2) \frac{X(z)}{z} \right] \Big|_{z=p_2}$$

$$C_3 = (z-p_2)^2 \frac{X(z)}{z} \Big|_{z=p_2}$$

Table 10.2 Inverse z -transform used for partial fraction expansion method.

S.No.	$X(z)$	$x(n) = Z^{-1}[X(z)]$ for ROC: $ z > a $
1.	$\frac{z}{z-a}$	$a^n u(n)$
2.	$\frac{z}{(z-a)^2}$	$na^{n-1} u(n)$
3.	$\frac{z}{(z-a)^3}$	$\frac{n(n-1)a^{n-2} u(n)}{2!}$
4.	$\frac{z}{(z-a)^4}$	$\frac{n(n-1)(n-2)}{3!} a^{n-3} u(n)$
$\vdots$	$\vdots$	$\vdots$
k	$\frac{z}{(z-a)^k}$	$\frac{n(n-1)\dots(n-(k-2))a^{n-k+1}}{(k-1)!} u(n)$

→ Find the inverse z-transform of the following :

$$i) \quad x(z) = \frac{\frac{1}{4}z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{4}z^{-1}\right)} \quad \text{ROC ; } |z| > \frac{1}{2}$$

A: Multiply by z^2

$$X(z) = \frac{\frac{1}{4}z}{\left(z - \frac{1}{2}\right)\left(z - \frac{1}{4}\right)}$$

$$\frac{X(z)}{z} = \frac{\frac{1}{4}}{\left(z - \frac{1}{2}\right)\left(z - \frac{1}{4}\right)} = \frac{A}{z - \frac{1}{2}} + \frac{B}{z - \frac{1}{4}}$$

$$A = \left(\cancel{z - \frac{1}{2}} \right) \frac{\frac{1}{4}}{\left(\cancel{z - \frac{1}{2}} \right) \left(z - \frac{1}{4} \right)} \bigg|_{z = \frac{1}{2}} = \frac{\frac{1}{4}}{\frac{1}{2} - \frac{1}{4}} = \frac{\frac{1}{4}}{\frac{2}{4}} = \underline{\underline{1}}$$

$$B = \left(\cancel{z - \frac{1}{4}} \right) \frac{\frac{1}{4}}{\left(z - \frac{1}{2} \right) \left(\cancel{z - \frac{1}{4}} \right)} \bigg|_{z = \frac{1}{4}} = \frac{\frac{1}{4}}{\frac{1}{4} - \frac{1}{2}} = \underline{\underline{-1}}$$

$$\frac{X(z)}{z} = \frac{1}{z - 1/2} - \frac{1}{z - 1/4}$$

$$X(z) = \frac{z}{z - 1/2} - \frac{z}{z - 1/4}$$

$$\text{ROC: } |z| > \frac{1}{2}$$

As ROC is $z > \frac{1}{2}$

seq. is causal.

$$x(n) = \left(\frac{1}{2}\right)^n u(n) - \left(\frac{1}{4}\right)^n u(n)$$

$$\rightarrow X(z) = \frac{8z - 19}{z^2 - 5z + 6}$$

$$\frac{X(z)}{z} = \frac{8z - 19}{z(z^2 - 5z + 6)}$$

$$= \frac{8z - 19}{z(z-3)(z-2)}$$

$$= \frac{A}{z} + \frac{B}{z-3} + \frac{C}{z-2}$$

$$A = \frac{8z - 19}{z(z-3)(z-2)} \cdot z \Big|_{z=0} = \frac{-19}{-3 \times -2} = \underline{\underline{-\frac{19}{6}}}$$

$$\left. \begin{array}{l} s = -5 \\ p = 6 \end{array} \right\} \begin{array}{l} 3 \\ 2 \end{array}$$

$$B = \frac{8z-19}{z(z-3)(z-2)} \times z-3 \Big|_{z=3}$$

$$= \frac{(8 \times 3) - 19}{3(3-2)}$$

$$= \frac{8}{3} \times \frac{5}{3}$$

$$C = \frac{8z-19}{z(z-3)(z-2)} \times (z-2) \Big|_{z=2}$$

$$= \frac{8 \times 2 - 19}{2(2-3)} = -\frac{3}{2} = \underline{\underline{\frac{3}{2}}}$$

$$\frac{x(z)}{z} = \frac{-19}{6z} + \frac{5}{3(z-3)} + \frac{3}{2(z-2)}$$

$$x(z) = -\frac{19}{6} + \frac{5z}{3(z-3)} + \frac{3}{2} \frac{z}{z-2}$$

$$x(n) = \frac{-19}{6} \delta(n) + \frac{5}{3} (3)^n u(n) + \underline{\underline{\frac{3}{2} (2)^n u(n)}}$$

$$\rightarrow X(z) = \frac{z(z-1)}{(z+2)^3(z+1)} \quad \text{ROC: } |z| > 2.$$

$$\frac{X(z)}{z} = \frac{z-1}{(z+2)^3(z+1)}$$

$$\frac{1}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} \left\{ (z-p)^n \frac{X(z)}{z} \right\}$$

$$= \frac{A}{z+2} + \frac{B}{(z+2)^2} + \frac{C}{(z+2)^3} + \frac{D}{z+1}$$

~~$$A = \frac{1}{2!} \frac{d^2}{dz^2} \left\{ (z+2)^3 \cdot \frac{z-1}{(z+2)^3(z+1)} \right\} \bigg|_{z=-2}$$~~

~~$$= \frac{z-1}{(z+2)^2(z+1)} \bigg|_{z=-2}$$~~

~~$$= \frac{-2-1}{-1} = \frac{3}{-1}$$~~

$$A = \frac{1}{(3-1)!} \frac{d^2}{dz^2} \left\{ (z+2)^3 \cdot \frac{z-1}{(z+2)^3(z+1)} \right\} \bigg|_{z=-2}$$

$$= \frac{1}{2!} \frac{d^2}{dz^2} \left[\frac{(z+1) \cdot 1 - (z-1)}{(z+1)^2} \right] \bigg|_{z=-2}$$

$$= \frac{1}{2!} \frac{d}{dz} \left[\frac{z+1 - z+1}{(z+1)^2} \right] \bigg|_{z=-2}$$

$$= \frac{1}{2} \left[\frac{(z+1) \cdot 0 - 1(z+1)}{(z+1)^4} \right] \Big|_{z=-2} \quad \begin{matrix} (z+1)^1 \\ 2(z+1) \end{matrix}$$

$$= \frac{1}{2} \left[\frac{-1(z+1)}{(z+1)^4} \right] \Big|_{z=-2}$$

$$= \frac{-1}{2} \left[\frac{-1}{(z+1)^3} \right] \Big|_{z=-2}$$

$$= \frac{-1}{2} \cdot \frac{+1}{(-2+1)^3} = \frac{-1}{2} \cdot \frac{4}{-1} = \underline{\underline{2}}$$

$$B = \frac{1}{1!} \frac{d}{dz} \left[(z+2)^3 \cdot \frac{z-1}{(z+2)^2(z+1)} \right] \Big|_{z=-2}$$

$$= \frac{(z+1) \cdot 1 - (z-1) \cdot 1}{(z+1)^2} \Big|_{z=-2}$$

$$= \frac{z+1 - z+1}{(z+1)^2} \Big|_{z=-2} = \frac{2}{(z+1)^2} \Big|_{z=-2} = \frac{2}{(-2+1)^2}$$

$$= \underline{\underline{2}}$$

$$C = (z+2)^3 \cdot \frac{z-1}{(z+2)^3(z+1)} \Big|_{z=-2}$$

$$= \frac{-2-1}{-2+1} = \frac{-3}{-1} = \underline{\underline{3}}$$

$$D = (z+1) \cdot \frac{z-1}{(z+2)^3(z+1)} \Big|_{z=-1}$$

$$= \frac{-1-1}{(-1+2)^3} = \underline{\underline{-2}}$$

$$\frac{x(z)}{z} = \frac{2}{z+2} + \frac{2}{(z+2)^2} + \frac{3}{(z+2)^3} - \frac{2}{z+1}$$

$$x(z) = \frac{2z}{z+2} + \frac{2z}{(z+2)^2} + \frac{3z}{(z+2)^3} + \frac{-2z}{z+1}$$

$$x(n) = 2(-2)^n u(n) + 2n(-2)^{n-1} u(n)$$

$$+ 3 \frac{n(n-1)(-2)^{n-2} u(n)}{2!} - 2(-1)^n u(n)$$

$$\rightarrow X(z) = \frac{8z - 19}{z^2 - 5z + 6}$$

$$\begin{aligned} \frac{X(z)}{z} &= \frac{8z - 19}{z(z^2 - 5z + 6)} \\ &= \frac{8z - 19}{z(z-2)(z-3)} \end{aligned}$$

$$= \frac{A}{z} + \frac{B}{z-2} + \frac{C}{z-3}$$

$$A = \left. \frac{8z - 19}{(z-2)(z-3)} \right|_{z=0} = \frac{-19}{-2 \times -3} = \underline{\underline{-\frac{19}{6}}}$$

$$B = \left. \frac{8z - 19}{z(z-3)} \right|_{z=2} = \frac{16 - 19}{2(-1)} = \underline{\underline{\frac{3}{2}}}$$

$$C = \left. \frac{8z - 19}{z(z-2)} \right|_{z=3} = \frac{8 \times 3 - 19}{3 \times 1} = \frac{5}{3}$$

$$\begin{aligned} X(z) &= -\frac{19}{6} \frac{z^1}{z} + \frac{3}{2} \frac{z}{z-2} + \frac{5}{3} \frac{z}{z-3} \\ &= -\frac{19}{6} \delta(n) + \frac{3}{2} (2)^n u(n) + \frac{5}{3} \underline{\underline{(3)^n u(n)}} \end{aligned}$$

Table 10.3 z-transform properties.

Property	Sequence	z-transform	ROC
	$x(n)$	$X(z)$	$R : R_1 < z < R_2$
	$x_1(n)$	$X_1(z)$	R_1
	$x_2(n)$	$X_2(z)$	R_2
Linearity	$a_1 x_1(n) + a_2 x_2(n)$	$a_1 X_1(z) + a_2 X_2(z)$	Intersection of R_1 and R_2
Time shifting	$x(n - m)$	$z^{-m} X(z)$	Same as $X(z)$ except $z = 0$ if $m > 0$ and $z = \infty$ if $m < 0$
Multiplication by exponential sequence	$a^n x(n)$	$X(a^{-1}z)$	$ a R_1 < z < a R_2$
Time reversal	$x(-n)$	$X(z^{-1})$	$\frac{1}{R_2} < z < \frac{1}{R_1}$
Differentiation in the z-domain	$nx(n)$	$-z \frac{d}{dz} X(z)$	$R_1 < z < R_2$
Conjugation	$x^*(n)$	$X^*(z^*)$	$R_1 < z < R_2$
Convolution	$x_1(n) * x_2(n)$	$X_1(z)X_2(z)$	At least the intersection of R_1 and R_2
Correlation	$x_1(l) * x_2(-l)$	$X_1(z)X_2(z^{-1})$	At least the intersection of ROC of $X_1(z)$ and $X_2(z^{-1})$
Multiplication	$x_1(n)x_2(n)$	$\frac{1}{2\pi j} \oint_c X_1(v)X_2\left(\frac{z}{v}\right)v^{-1}dv$	At least $R_{1l}R_{2l} < z < R_{1u}R_{2u}$
Initial value theorem	$x(0) = \lim_{z \rightarrow \infty} X(z)$		
Final value theorem	$x(\infty) = \lim_{z \rightarrow 1} (1 - z^{-1})X(z)$, if $(1 - z^{-1})X(z)$ has no pole on or outside the unit circle.		
Parseval's relation	$\sum_{n=-\infty}^{\infty} x_1(n)x_2^*(n) = \frac{1}{2\pi j} \oint_c X_1(v)X_2^*\left(\frac{1}{v^*}\right)v^{-1}dv$		

10.6.3 Residue method

The inverse z -transform of $X(z)$ can be obtained using Eq. (10.8) that is

$$x(n) = \frac{1}{2\pi j} \oint_c X(z) z^{n-1} dz \quad (10.85)$$

where c is a circle in the z -plane in the ROC of $X(z)$.

The Eq. (10.85) can be evaluated by finding sum of all residues of the poles that are inside the circle c . Therefore Eq. (10.85) can be rewritten as

$$\begin{aligned} x(n) &= \sum (\text{residues of } X(z)z^{n-1} \text{ at the poles inside } c) \\ &= \sum_i (z - z_i) X(z) z^{n-1} \Big|_{z=z_i} \end{aligned} \quad (10.86)$$

If $X(z)z^{n-1}$ has no poles inside the contour c for one or more values of n , then $x(n) = 0$ for these values. The residues of the poles can be found by using cauchy residue theorem.

Cauchy residue theorem

Let $f(z)$ be a function of the complex variable z and c be a closed path in the z -plane. If the derivative $\frac{d}{dz} f(z)$ exists on and inside the contour c and if $f(z)$ has no poles at $z = z_0$, then

$$\frac{1}{2\pi j} \oint_c \frac{f(z)}{z - z_0} dz = f(z_0) \text{ if } z_0 \text{ is inside } c$$
$$= 0 \text{ if } z_0 \text{ is outside } c \quad (10.87)$$

If the $(k + 1)$ order derivative of $f(z)$ exists and $f(z)$ has no poles at $z = z_0$, then

$$\frac{1}{2\pi j} \oint_c \frac{f(z)}{(z - z_0)^k} dz = \frac{1}{(k - 1)!} \frac{d^{k-1}}{dz^{k-1}} f(z) \Big|_{z=z_0} \text{ if } z_0 \text{ is inside } c \quad (10.88)$$
$$= 0 \text{ if } z_0 \text{ is outside } c$$

The values on the right hand side of Eq. (10.87) and Eq. (10.88) are called residues of the pole at $z = z_0$. If there are more than one pole inside the contour, the integral can be evaluated by adding up the results from each pole. For example if there are n unique poles $z_1, z_2, \dots, z_n$ and $f(z)$ containing no poles inside c , then

$$\frac{1}{2\pi j} \oint_c \frac{f(z) dz}{(z - z_1)(z - z_2) \dots (z - z_n)} = \sum_{i=1}^n \lim_{z \rightarrow z_i} [(z - z_i) \gamma(z)] \quad (10.89)$$

where

$$\gamma(z) = \frac{f(z)}{(z - z_1)(z - z_2) \dots (z - z_n)} \quad (10.90)$$

If any of poles are of multiplicity greater than one, the formula given in Eq. (10.88) is used to determine their contribution to the integral, while Eq. (10.87) is used for simple poles.

Example 10.24: Using residue method find the inverse z -transform of

$$X(z) = \frac{1 + 3z^{-1}}{1 + 3z^{-1} + 2z^{-2}} \quad |z| > 2$$

Solution Given

$$X(z) = \frac{1 + 3z^{-1}}{1 + 3z^{-1} + 2z^{-2}} = \frac{z(z + 3)}{z^2 + 3z + 2} = \frac{z(z + 3)}{(z + 1)(z + 2)}$$

from Eq. (10.85)

$$\begin{aligned} x(n) &= \frac{1}{2\pi j} \oint_c X(z) z^{n-1} dz \\ &= \sum \text{residues of } X(z) z^{n-1} \text{ at poles of } X(z) z^{n-1} \text{ within } c \\ &= \sum \text{residues of } \frac{z(z+3)}{(z+1)(z+2)} z^{n-1} \text{ at poles of same within } c \end{aligned}$$

The closed contour c , being in the ROC $|z| > 2$ encloses the poles at $z = -1$ and $z = -2$

$$\begin{aligned} x(n) &= \sum \text{residues of } \frac{(z+3)z^n}{(z+1)(z+2)} \text{ at poles } z = -1 \text{ and } z = -2 \\ &= \text{residue of } \frac{(z+3)z^n}{(z+1)(z+2)} \text{ at } z = -1 + \text{residue of } \frac{(z+3)z^n}{(z+1)(z+2)} \text{ at } z = -2 \\ &= \left. \frac{(z+1)}{(z+1)(z+2)} \frac{(z+3)z^n}{(z+1)(z+2)} \right|_{z=-1} + \left. \frac{(z+2)}{(z+1)(z+2)} \frac{(z+3)z^n}{(z+1)(z+2)} \right|_{z=-2} \\ &= 2(-1)^n - (-2)^n \end{aligned}$$

For $n < 0$ we can prove $x(n) = 0$.

For $n = -1$

$$\begin{aligned} X(z) z^{n-1} &= \frac{z+3}{z(z+1)(z+2)} \\ x(-1) &= \sum \text{residues of } \frac{z+3}{z(z+1)(z+2)} \text{ at } z = 0, -1 \text{ and } -2 \\ &= \text{residue of } \frac{z+3}{z(z+1)(z+2)} \text{ at } z=0 + \text{residue of } \frac{z+3}{z(z+1)(z+2)} \text{ at } z=-1 \\ &\quad + \text{residue of } \frac{z+3}{z(z+1)(z+2)} \text{ at } z=-2 \\ &= \left. \frac{z+3}{z(z+1)(z+2)} \right|_{z=0} + \left. \frac{(z+1)}{z(z+1)(z+2)} \frac{z+3}{(z+1)(z+2)} \right|_{z=-1} \\ &\quad + \left. \frac{(z+2)}{z(z+1)(z+2)} \frac{z+3}{(z+1)(z+2)} \right|_{z=-2} \\ &= \frac{3}{2} - 2 + \frac{1}{2} = 0. \end{aligned}$$

Similarly we can prove for other negative values of n .

Therefore for $n < 0$ $x(n) = 0$

$$x(n) = 2(-1)^n u(n) - (-2)^n u(n)$$

Example 10.25: Determine the inverse z -transform using the complex integral

$$(a) \quad X(z) = \frac{2z^{-1}}{\left(1 - \frac{1}{4}z^{-1}\right)^2} \quad \text{ROC: } |z| > \frac{1}{4}$$

$$(b) \quad X(z) = \frac{z^{-1}}{(1 - 2z^{-1})(1 - 3z^{-1})} \quad \text{ROC: } 2 < |z| < 3$$

Solution

$$(a) \quad \text{Given } X(z) = \frac{2z^{-1}}{\left(1 - \frac{1}{4}z^{-1}\right)^2} \\ = \frac{2z}{\left(z - \frac{1}{4}\right)^2}$$

Using complex integral

$$x(n) = \frac{1}{2\pi j} \oint_c X(z) z^{n-1} dz$$

For $n \geq 0$

$$x(n) = \frac{1}{2\pi j} \oint_c \frac{2z}{\left(z - \frac{1}{4}\right)^2} \cdot z^{n-1} dz \\ = \frac{1}{2\pi j} \oint_c \frac{2z^n}{\left(z - \frac{1}{4}\right)^2} dz \\ = \text{residue of } \frac{2z^n}{\left(z - \frac{1}{4}\right)^2} \text{ at the pole } z = \frac{1}{4} \text{ of multiplicity 2}$$

Therefore using Eq. (10.88) we have

$$x(n) = \frac{1}{1!} \frac{d}{dz} \left[\left(z - \frac{1}{4}\right)^2 \frac{2z^n}{\left(z - \frac{1}{4}\right)^2} \right] \bigg|_{z=\frac{1}{4}} \\ = 2n(z)^{n-1} \bigg|_{z=\frac{1}{4}} \\ = 2n \left(\frac{1}{4}\right)^{n-1}$$

10.48 Signals and Systems

For $n = -1$

$$\begin{aligned} x(-1) &= \frac{1}{2\pi j} \oint_c \frac{1}{z \left(z - \frac{1}{4}\right)^2} dz \\ &= \left. z \cdot \frac{1}{\left(z - \frac{1}{4}\right)^2} \right|_{z=0} + \frac{1}{1!} \frac{d}{dz} \left[\left(z - \frac{1}{4}\right)^2 \cdot \frac{1}{z \left(z - \frac{1}{4}\right)^2} \right] \bigg|_{z=\frac{1}{4}} \\ &= 16 - 16 = 0 \end{aligned}$$

i.e., for $n < 0$ $x(n) = 0$
Therefore

$$x(n) = 2n \left(\frac{1}{4}\right)^{n-1} u(n)$$

(b) Given

$$\begin{aligned} X(z) &= \frac{z^{-1}}{(1 - 2z^{-1})(1 - 3z^{-1})} \quad \text{ROC: } 2 < |z| < 3 \\ &= \frac{z}{(z - 2)(z - 3)} \end{aligned}$$

The contour of integration c lies in the annular region of ROC, and the inverse z -transform is

$$\begin{aligned} x(n) &= - \sum \text{residue of } X(z)z^{n-1} \text{ at pole } z = 3 \text{ for } n < 0 \\ &= \sum \text{residue of } X(z)z^{n-1} \text{ at pole } z = 2 \text{ for } n \geq 0 \end{aligned}$$

For $n < 0$

$$\begin{aligned} x(n) &= \left. \frac{z^n}{(z - 2)(z - 3)} \right|_{z=3} \\ &= -(3)^n \end{aligned}$$

For $n \geq 0$

$$\begin{aligned} x(n) &= \left. \frac{(z - 2) z^n}{(z - 2)(z - 3)} \right|_{z=2} \\ &= -(2)^n \end{aligned}$$

Therefore

$$x(n) = -(2)^n u(n) - (3)^n u(-n - 1)$$

Exercise 10.14:

Find the inverse z-transform of the following using Residue method

$$(i) \quad X_1(z) = \frac{z^2 + z}{(z-1)(z-3)} \quad \text{ROC: } |z| > 3 \quad \text{Ans:}$$

$$(ii) \quad X_2(z) = \frac{1 - \frac{1}{4}z^{-1}}{1 - \frac{1}{9}z^{-1}} \quad \text{ROC: } |z| > \frac{1}{3}$$

$$(iii) \quad X_3(z) = \frac{z^2}{\left(z - \frac{1}{4}\right)^2} \quad \text{ROC: } |z| > \frac{1}{4}$$

$$(i) \quad 2(3)^n u(n) - u(n)$$

$$(ii) \quad \frac{1}{8} \left(\frac{1}{3}\right)^n + \frac{7}{8} \left(-\frac{1}{3}\right)^n u(n).$$

$$(iii) \quad (n+1) \left(\frac{1}{4}\right)^n u(n)$$

10.6.4 Convolution method

In this method the given $X(z)$ is splitted into $X_1(z)$ and $X_2(z)$ such that $X(z) = X_1(z)X_2(z)$. Next we find $x_1(n)$ and $x_2(n)$ by taking inverse z- transform of $X_1(z)$ and $X_2(z)$ respectively.

From convolution property of z- transform we know

$$Z[x_1(n) * x_2(n)] = X_1(z)X_2(z) = X(z) \quad (10.91)$$

From Eq. (10.91) we know that the convolution of $x_1(n)$ and $x_2(n)$ is the inverse z- transform of $X(z)$. Thus $x(n)$ can be obtained by convolving $x_1(n)$ and $x_2(n)$.

Example 10.26: Repeat the Example 10.24 using convolution method.

Solution

$$\begin{aligned} \text{Given } X(z) &= \frac{1 + 3z^{-1}}{1 + 3z^{-1} + 2z^{-2}}; \quad |z| > 2 \\ &= \frac{z(z+3)}{(z+1)(z+2)} \end{aligned}$$

$$\text{let } X(z) = X_1(z)X_2(z)$$

$$\text{where } X_1(z) = \frac{z}{z+1}$$

$$X_2(z) = \frac{z+3}{z+2} = \frac{z}{z+2} + z^{-1} \frac{3z}{z+2}$$

$$\text{Then } x_1(n) = (-1)^n u(n) \quad \text{and}$$

$$x_2(n) = (-2)^n u(n) + 3(-2)^{n-1} u(n-1)$$

10.50 Signals and Systems

$$\begin{aligned}
 x(n) &= x_1(n) * x_2(n) \\
 &= (-1)^n u(n) * \left[(-2)^n u(n) + 3(-2)^{n-1} u(n-1) \right] \\
 &= (-1)^n u(n) * (-2)^n u(n) + 3(-1)^n u(n) * (-2)^{n-1} u(n-1) \\
 &= \sum_{k=0}^n (-1)^k u(k) (-2)^{n-k} u(n-k) \\
 &\quad + 3 \sum_{k=0}^{n-1} (-1)^k u(k) (-2)^{n-k-1} u(n-k-1) \\
 &= (-2)^n \sum_{k=0}^n (-1)^k (-2)^{-k} + 3(-2)^{n-1} \sum_{k=0}^{n-1} (-1)^k (-2)^{-k} \\
 &= (-2)^n \sum_{k=0}^n (0.5)^k + 3(-2)^{n-1} \sum_{k=0}^{n-1} (0.5)^k \\
 &= (-2)^n \left[\frac{1 - (0.5)^{n+1}}{1 - 0.5} \right] + 3(-2)^{n-1} \left[\frac{1 - (0.5)^n}{1 - 0.5} \right] \\
 &= (-2)^n \cdot 2[1 - 0.5(0.5)^n] + 3(-2)^{n-1} \cdot 2[1 - (0.5)^n] \\
 &= (-2)^n 2[1 - 0.5(0.5)^n] - 3(-2)^n [1 - (0.5)^n] \\
 &= 2(-2)^n - (-1)^n - 3(-2)^n + 3(-1)^n \\
 &= 2(-1)^n - (-2)^n \quad \text{for } n \geq 0 \\
 x(n) &= [2(-1)^n - (-2)^n] u(n)
 \end{aligned}$$

$$\boxed{\sum_{k=0}^n a^k = \frac{1 - a^{n+1}}{1 - a}}$$